

Continuous Cash-Overlay Filters for a Static Growth–Defensive Risk Sleeve

Direct Slow-Tail Compensation, Fixed V-Shape Crash Brakes, Walk-Forward Validation, and Max-Cash Combination

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Code repository: <https://github.com/shaun19920309/gd-cash-overlay-filters>

Abstract

This paper studies a modular cash-overlay rule for allocating between a fixed growth–defensive risky sleeve R and interest-bearing cash C . The risky sleeve is a static 50/50 combination of equal-weight growth/technology and defensive income/value ETF baskets; the target is future $R - C$ return with the cash leg earning the contemporaneous cash rate. Two independent filters are tested. The slow-tail filter maps continuous compensation, rate-headwind, risk-premium-compression, and rate-path-stress states into a cash weight with a 30% material-trade gate. The V-shape filter is a fast crash brake based on continuous VIX, rate, credit, drawdown, and re-entry states. A fixed max-cash layer then uses the larger cash weight requested by either filter each day. On the 2017–2026 common window, the selected max-cash combination earns an 18.83% CAGR versus 16.62% for 100% R and reduces maximum drawdown from -33.59% to -18.05%. In the main walk-forward OOS window, the expanding combination earns 19.35% versus 17.59% for 100% R , with maximum drawdown of -22.05% versus -33.59%; the rolling version earns 18.50% with the same -22.05% drawdown. Post-2022 tests show lower drawdown but lower CAGR during a strong risky-sleeve rebound. The results support modular cash overlays as drawdown-control tools rather than standalone return-enhancement claims; fully real-time variable re-screening and multiple-testing-adjusted inference remain future work.

Keywords: dynamic asset allocation; cash overlay; crash protection; VIX; interest rates; credit stress; walk-forward validation; drawdown control.

1 Introduction

This paper asks whether a static growth–defensive risky sleeve should sometimes be replaced by interest-bearing cash. The problem is deliberately narrower than a full asset-allocation model. The risky sleeve R is fixed, transparent, and independent of any dynamic growth–defensive timing rule. The overlay only decides how much of this risky sleeve to hold versus cash.

The research design is modular, but the two modules do not change equally in this version. The V-shape branch is retained as a fixed crash brake for fast drawdowns. The slow-tail branch is rebuilt as a direct compensation score rather than a historical analogue estimator. The two filters are then combined with a conservative max-cash rule. This design avoids a single broad regime classifier and instead treats persistent compensation deterioration and fast crash risk as different alarms.

The paper follows an empirical strategy-study structure. It first defines the static risky sleeve and the cash benchmark. It then documents the statistical screening logic, the two policy rules, the parameter grids, full-sample selected results, walk-forward validation, event diagnostics, and the final max-cash combination. Appendix tables preserve the main screening and robustness outputs.

2 Related Literature

The study sits between return-predictability research, parametric portfolio policy design, and practical crash-risk management. The predictive-return literature warns that variables with in-sample explanatory power often fail in out-of-sample settings, especially when candidate variables are numerous and labels overlap [Goyal and Welch(2008), Campbell and Thompson(2008)]. This motivates the paper’s separation between diagnostic screening and trading evidence: Newey–West regressions, non-overlapping labels, and multi-horizon consistency are used to organize candidate state variables, while walk-forward portfolio tests decide whether a component is useful as a cash overlay.

The policy design is closest in spirit to parametric allocation rules. Brandt, Santa-Clara, and Valkanov [Brandt, Santa-Clara, and Valkanov(2009)] show how portfolio weights can be linked directly to state variables. Instead of estimating an unconstrained mean-variance optimizer, this paper maps economically interpretable state variables into a continuous cash weight. This keeps the action space narrow: the risky sleeve is fixed, and the policy can only reduce or restore exposure to that sleeve. The material cash gate of 30% is not a predictive threshold. It is an implementation rule that says a computed cash allocation must be large enough to matter before a trade is taken.

The robustness discussion follows the data-snooping literature. Full-sample screening and policy construction are useful for discovering economically coherent mechanisms, but they do not by themselves establish a final anomaly claim. White’s Reality Check, Hansen’s SPA test, and the Deflated Sharpe Ratio are therefore treated as future confirmation tools rather than as claims already completed in this version [White(2000), Hansen(2005), Bailey and Lopez de Prado(2014)]. The validation discipline follows the companion growth–defensive style-timing study [Xiong(2026)]: first build a transparent empirical mechanism, then test whether it survives increasingly strict validation layers.

3 Static Risky Sleeve and Cash Return

Let G_t denote the equal-weight return of the growth/technology basket {QQQ, XLK, VGT, SPYG, VUG}. Let D_t denote the equal-weight return of the defensive income/value

basket $\{\text{SCHD}, \text{VYM}, \text{VTV}, \text{FDVV}, \text{COWZ}\}$. The risky sleeve is

$$R_t^R = 0.5G_t + 0.5D_t. \quad (1)$$

Because each basket contains five ETFs, this is equivalent to a 10-ETF equal-weight risky sleeve. The cash sleeve return R_t^C is constructed from the contemporaneous cash yield. The target excess return for screening is

$$R_t^{R-C} = R_t^R - R_t^C. \quad (2)$$

This construction is intentionally independent of the dynamic growth–defensive style-allocation policy studied in [Xiong(2026)]. That companion policy may later allocate within the risky sleeve between growth and defensive assets, but the cash-overlay experiments in this paper do not use its scores, states, or weights. Throughout the paper, “100% R ” means the static 50/50 G/D risky sleeve, not a dynamic style-timing policy.

Table 1: Data Coverage and Portfolio Definitions

Object	Definition	Sample or Role
Growth basket G	Equal-weight QQQ, XLK, VGT, SPYG, VUG	Technology and growth equity exposure.
Defensive basket D	Equal-weight SCHD, VYM, VTV, FDVV, COWZ	Dividend, value, quality, and cash-flow tilted equity exposure.
Risky sleeve R	$0.5G + 0.5D$	Static 10-ETF equal-weight risky pool used in every cash-overlay comparison.
Cash sleeve C	Contemporaneous cash/risk-free return	Interest-bearing cash benchmark; cash yield enters both returns and state variables.
Feature panel	2017-06-28–2026-04-30, 2222 trading days	Shared state panel used by slow-tail and V-shape modules.
Main OOS window	2018-06-28 requested start, realized from first valid 63-day test block	Walk-forward parameter re-selection.
Post-2022 OOS window	2022-01-03–2026-04-30	Stress test for the slow-tail regime and the post-rate-hike environment.

The final common-window comparisons apply a 10bp one-way transaction cost unless otherwise stated. Turnover is measured as the one-way change in the risky weight, $|w_t^R - w_{t-1}^R|$, whenever selected component weights and max-cash combined weights are recomputed on the common daily panel. Component-native diagnostic tables retain the archived module convention used by the corresponding component script. This matters mainly for the V-shape standalone tables, whose native turnover counts both risky-weight and cash-weight changes; the max-cash section therefore reports a common-window V-shape-only recomputation before combining the modules.

3.1 Raw State Variables

The state panel is built from observable prices, yields, and spread proxies. ETF returns use adjusted daily close data. Treasury and credit-yield variables enter in percentage-point units before standardization. VIX variables enter either as rolling percentiles or percentage changes, and credit-risk proxies use both cash-bond yield spreads and ETF relative returns. The risky-sleeve variables are rebuilt from the static R definition in this paper, so no companion style-timing score, state, or dynamic G/D weight enters the cash-overlay component panel.

Table 2: Raw State Variables Used in Cash-Overlay Components

Variable	Construction	Component Role
R_t^R	$0.5G_t + 0.5D_t$, where G and D are equal-weight ETF baskets.	Risky-sleeve return used in both strategy returns and all R -dependent state variables.
R_t^C	Daily interest-bearing cash return from the contemporaneous cash/risk-free series.	Cash benchmark return; $R - C$ is the screening target.
Cash yield	Annualized cash yield, with 21-day change.	Cash compensation level and cash-rising pressure.
10-year Treasury yield	21-day change in TNX.	Rate headwind in slow-tail; rate relief in V-shape after sign reversal.
10-year minus 3-month curve	TNX minus IRX.	Curve-inversion diagnostic in the main-effect screen.
R drawdown	Cumulative R wealth divided by its running peak minus one.	Drawdown depth and oversold state.
R trailing return	10-, 21-, 63-, 126-, and 252-day cumulative R return, with the 10-day version used only in V-shape crash-loss interactions.	Risky-sleeve strength, underwater state, and short-horizon crash loss.
R volatility	63-day realized volatility of R , annualized by $\sqrt{252}$.	Low-volatility compensation state.
VIX level	756-day rolling percentile of VIX.	High/low volatility state and V-shape level core.
VIX changes	21-day VIX change for slow-tail; 10-day VIX change and 5-day VIX relief for V-shape.	Volatility spike and volatility relief.
HYG-SHY relative return	21-day HYG return minus 21-day SHY return; 5-day version is used for V-shape credit appetite.	Credit risk-off and credit appetite.
BAA-AAA spread	Moody's BAA yield minus AAA yield, and its 21-day change.	Credit widening and panic interaction.

All continuous states are transformed with expanding standardization:

$$z_t(x) = \frac{x_t - \mu_t^{exp}(x)}{\sigma_t^{exp}(x)}. \quad (3)$$

For direct slow-tail components, the expanding window uses a 252-trading-day warmup and a one-day lag before a standardized value is used. V-shape primitives use the same expanding-standardization idea, with the candidate score layer standardized again before the deployable brake score is formed. The current V-shape branch is retained from the fixed material-brake version; the methodological update in this paper is the direct slow-tail compensation score.

4 Statistical Screening and Validation Design

Directional intensities use smooth transformations such as $softplus(x) = \log(1 + \exp(x))$. The retained filters do not use binary feature truncation. The only hard trading gate is economic materiality: a cash move is implemented only when the continuously computed raw cash weight exceeds 30%.

Candidate variables are screened against future $R - C$ returns at 21, 63, and 126 trading-day horizons. Because labels overlap, the primary regression diagnostic is the OLS coefficient with Newey–West/HAC standard errors. The screen also reports non-overlapping 63-day regressions, multi-horizon consistency, economic sign, and coverage. These tests are used to organize candidate components. Trading evidence is evaluated separately through full-sample grid selection and walk-forward parameter re-selection.

For each strategy, the full-sample grid is a descriptive fitted result. In OOS tests, each 63-trading-day test block re-selects the complete parameter vector using either all prior data (expanding) or the most recent 756 trading days (rolling). Thus, OOS return series are not fixed-parameter backtests. Appendix Tables 32 and 33 report the complete OOS return matrix and the block-level parameter-selection diagnostics.

5 Baseline Annual Diagnostics

Before evaluating timing rules, it is important to separate three facts. First, the static risky sleeve has a high unconditional return over the sample, so a cash overlay must preserve most upside participation. Second, cash becomes economically relevant only after rate normalization; it cannot explain the 2020 crash-protection result. Third, the two modules add value in different years: the V-shape module matters most in 2020 and 2025, while the slow-tail module matters most in 2022.

The annual decomposition makes the modular nature of the overlay visible. In Table 3, the combined policy is not uniformly more aggressive than its components. It matches the V-shape module in 2020, matches the slow-tail module in 2022, and accepts modest opportunity cost in calm years when a component remains partially in cash.

Table 3: Annual Return / Maximum Drawdown by Policy, Selected-Weight Common Window

Year	100% R		100% C		Slow-tail		V-shape		Max-cash	Avg.	Max-cash	Cash
2018	-3.02%	/ -19.71%	2.54%	/ 0.00%	-3.02%	/ -19.71%	-2.31%	/ -16.77%	-2.31%	/ -16.77%		7.68%
2019	32.90%	/ -8.03%	2.55%	/ 0.00%	32.90%	/ -8.03%	32.39%	/ -8.03%	32.39%	/ -8.03%		0.57%
2020	23.82%	/ -33.59%	0.62%	/ 0.00%	23.82%	/ -33.59%	44.55%	/ -15.11%	44.55%	/ -15.11%		7.64%
2021	31.04%	/ -4.92%	0.00%	/ 0.00%	30.66%	/ -4.92%	29.36%	/ -4.92%	28.99%	/ -4.92%		1.30%
2022	-17.06%	/ -23.78%	1.29%	/ 0.00%	-8.95%	/ -18.05%	-17.06%	/ -23.77%	-8.95%	/ -18.05%		15.60%
2023	28.14%	/ -9.73%	5.13%	/ 0.00%	27.80%	/ -9.73%	28.14%	/ -9.73%	27.80%	/ -9.73%		0.56%
2024	22.51%	/ -8.88%	5.17%	/ 0.00%	22.51%	/ -8.88%	18.60%	/ -7.93%	18.60%	/ -7.93%		3.28%
2025	17.31%	/ -19.66%	5.13%	/ 0.00%	17.31%	/ -19.66%	15.79%	/ -14.64%	15.79%	/ -14.64%		7.02%
2026	8.61%	/ -7.95%	1.03%	/ 0.00%	7.32%	/ -7.81%	6.64%	/ -7.86%	6.80%	/ -7.73%		7.97%

6 Policy Grids and Selection Objectives

Both modules select score-construction parameters and position-mapping parameters. The direct slow-tail grid has 216 configurations:

$$216 = 2 \alpha_{comp} \times 2 \lambda_{rate} \times 2 \lambda_{path} \times 3 \max \text{ cash} \times 3 \tau \times 3 \text{ convexities}. \quad (4)$$

The V-shape grid has 216 configurations:

$$216 = 2 \alpha_{vix} \times 2 \lambda_{rate} \times 2 \lambda_{credit} \times 3 \max \text{ cash} \times 3 \tau \times 3 \text{ convexities}. \quad (5)$$

The material threshold is fixed at 30% in both grids. Entry to cash is immediate for V-shape and economically immediate for slow-tail because the target cash weight is used once it is material. Re-entry to R is smoothed with $\eta_{exit} = 0.25$.

The grid objective is deliberately multi-criteria. A pure CAGR sort would choose overly aggressive rules that fail the risk-management purpose of the cash overlay. A

pure drawdown sort would over-allocate to cash. The selection score ranks configurations by return, drawdown, Calmar, event performance, turnover, and average cash use. The exact weights differ by module because the slow-tail module is a compensation filter and the V-shape module is a crash brake, but both penalize high turnover and unnecessary cash exposure.

Table 4: Parameter Grid Design

Module	Grid Parameters	Interpretation
Slow-tail	$\alpha_{comp} \in \{0.50, 0.67\}$; $\lambda_{rate} \in \{0.25, 0.50\}$; $\lambda_{path} \in \{0.25, 0.50\}$; max cash $\in \{0.50, 0.75, 1.00\}$; cash $\tau \in \{0.75, 1.00, 1.25\}$; convexity $\in \{2, 3, 4\}$.	Chooses the weight on cash-rising pressure, compressed-risk-premium components, and rate-path stress, then maps the direct score into a cash weight.
V-shape	$\alpha_{vix} \in \{0.50, 0.67\}$; $\lambda_{rate} \in \{0.25, 0.50\}$; $\lambda_{credit} \in \{0, 0.25\}$; max cash $\in \{0.50, 0.75, 1.00\}$; brake $\tau \in \{0.50, 0.75, 1.00\}$; convexity $\in \{3, 4, 5\}$.	Chooses the mix of VIX level/spike, rate panic, and credit panic, then maps the crash-brake score into a cash weight.
Max-cash combination	No new fitted parameter.	Uses $w_t^C = \max(w_t^{C,slow}, w_t^{C,vshape})$; when both modules are evaluated OOS, the combination uses their OOS-selected weights, not full-sample component weights.

Table 5: Selected Full-Sample Results, 10bp Cost

Policy	Window	CAGR	Sharpe	Max DD	Ann. Turnover	Avg. Cash	100% R CAGR	100% R Max DD
Slow-tail	2018-06-28–2026-04-30	17.83%	0.94	-33.59%	108.36%	2.19%	16.69%	-33.59%
V-shape	2017-07-19–2026-04-30	17.32%	1.07	-23.77%	532.41%	3.65%	16.62%	-33.59%
Max-cash combination	2017-07-19–2026-04-30	18.83%	1.18	-18.05%	355.82%	5.56%	16.62%	-33.59%

The selected-weight comparison is best read as a fitted economic map rather than as the main OOS claim. Table 5 shows the role each module is designed to play before moving to walk-forward tests. The redesigned slow-tail filter raises CAGR modestly and improves 2022, but it leaves the full-sample maximum drawdown unchanged because it is not a fast-crash tool. The fixed V-shape branch gives a smaller CAGR gain but materially reduces maximum drawdown. The max-cash combination improves both dimensions because it inherits slow-tail protection in 2022 and V-shape protection in fast crashes. Later combination tables recompute the same selected component weights on a common panel with the final one-way turnover convention, so the V-shape-only row in Table 13 is intentionally not identical to the standalone V-shape row in this summary table.

7 Slow-Tail Compensation Filter

7.1 Policy Construction

The slow-tail filter is the part of the paper that changes materially in the current experiment. The previous historical-analogue construction is replaced by a direct continuous compensation score. The goal is still the same: identify persistent states in which holding R over interest-bearing cash is poorly compensated. The implementation, however, no longer estimates a tail historical future-return mean. It combines screened continuous components into a direct score and then maps that score into a raw cash weight.

The signed primitive variables are continuous expanding z -scores. The cash-rising component is

$$CashRise_t = EZ(\Delta CashYield_{21,t}). \quad (6)$$

The compressed-risk-premium block averages five screened low-compensation components:

$$LCP_t = \frac{1}{5} (EZ(RS_t LVOL_t) + EZ(RS_t LVOL_t LV_t) + EZ(RS_t LV_t) + EZ(CH_t RS_t LV_t) + EZ(CH_t RS_t LVOL_t)), \quad (7)$$

where RS is risky-sleeve strength, $LVOL$ is low realized risky-sleeve volatility, LV is low VIX, and CH is high cash yield. These products describe states in which the risky sleeve has been strong and calm while cash is becoming more competitive.

The compensation core is

$$CompCore_t = \alpha_{comp} CashRise_t + (1 - \alpha_{comp}) LCP_t. \quad (8)$$

The direct rate headwind is $RH_t = EZ(\Delta TNX_{21,t})$. The rate-path stress block averages four screened interactions:

$$RPS_t = \frac{1}{4} (EZ(RH_t VS_t) + EZ(RH_t CRO_t) + EZ(RH_t RU_t) + EZ(DD_t RH_t CRO_t)), \quad (9)$$

where VS is a VIX spike, CRO is credit risk-off, RU is risky-sleeve underwater, and DD is drawdown depth. The final slow-tail score is

$$SlowScore_t = CompCore_t + \lambda_{rate} RH_t + \lambda_{path} RPS_t, \quad SlowZ_t = EZ(SlowScore_t). \quad (10)$$

The raw cash weight is continuous:

$$CashRaw_t = MaxCash \left[\frac{1}{1 + \exp(-SlowZ_t/\tau)} \right]^\gamma. \quad (11)$$

If $CashRaw_t \geq 0.30$, entry into cash is immediate. If $CashRaw_t < 0.30$, the strategy smoothly returns toward R with an exit smoothing speed of 0.25. The selected full-sample configuration is **direct_slow_187**. The grid contains 216 configurations over α_{comp} , λ_{rate} , λ_{path} , max cash, cash scale, and cash convexity.

Table 6: Slow-Tail Top Configurations by Selection Score

Config	α_{comp}	λ_{rate}	λ_{path}	Max Cash	τ	Power	CAGR	Score
direct_slow_187	0.67	0.50	0.25	1.00	1.25	2	17.83%	0.740
direct_slow_184	0.67	0.50	0.25	1.00	1.00	2	17.60%	0.733
direct_slow_214	0.67	0.50	0.50	1.00	1.25	2	17.51%	0.732
direct_slow_157	0.67	0.25	0.50	1.00	1.00	2	17.36%	0.729
direct_slow_079	0.50	0.50	0.25	1.00	1.25	2	17.35%	0.724
direct_slow_127	0.67	0.25	0.25	1.00	0.75	2	17.32%	0.722
direct_slow_118	0.67	0.25	0.25	0.75	0.75	2	17.20%	0.722
direct_slow_130	0.67	0.25	0.25	1.00	1.00	2	17.25%	0.721

The selected slow-tail structure is not an isolated single parameter point. The leading configurations cluster around $\alpha_{comp} = 0.67$, high max cash, low convexity, and positive

weights on both rate headwind and rate-path stress (Table 6). The implication is that current slow-tail performance is driven by the direct compensation construction rather than by the old analogue-tail parameters.

Table 7: Slow-Tail Standalone Walk-Forward Validation

Filter	Mode	CAGR	Max DD	Ann. Turnover	Avg. Cash	100% R CAGR	100% R Max DD
Slow-tail	Expanding	18.79%	-33.59%	114.57%	2.44%	17.59%	-33.59%
Slow-tail	Rolling	18.12%	-33.59%	138.31%	2.77%	17.59%	-33.59%

The first direct walk-forward check supports the slow-tail mechanism but also exposes its boundary. In Table 7, expanding and rolling validation both improve CAGR relative to 100% R , yet neither improves the full-window maximum drawdown because the worst drawdown in the main window is a fast crash. This is exactly the boundary of the module: it is a compensation filter for persistent weak $R - C$ regimes, not a crash brake.

Table 8: Slow-Tail Post-2022 Walk-Forward Validation

Filter	Mode	CAGR	Max DD	Ann. Turnover	Avg. Cash	100% R CAGR	100% R Max DD
Slow-tail	Expanding	21.19%	-19.74%	169.65%	1.62%	23.58%	-19.66%
Slow-tail	Rolling	21.19%	-19.74%	169.65%	1.62%	23.58%	-19.66%

For slow-tail, the post-2022 test now shows a different trade-off from the previous analogue version. Both expanding and rolling validation hold little average cash and slightly trail 100% R on CAGR, while drawdown is essentially unchanged (Table 8). This weaker post-2022 standalone result is why the current paper treats the redesigned slow-tail branch as a modest 2022 compensation filter rather than as a complete slow-regime solution.

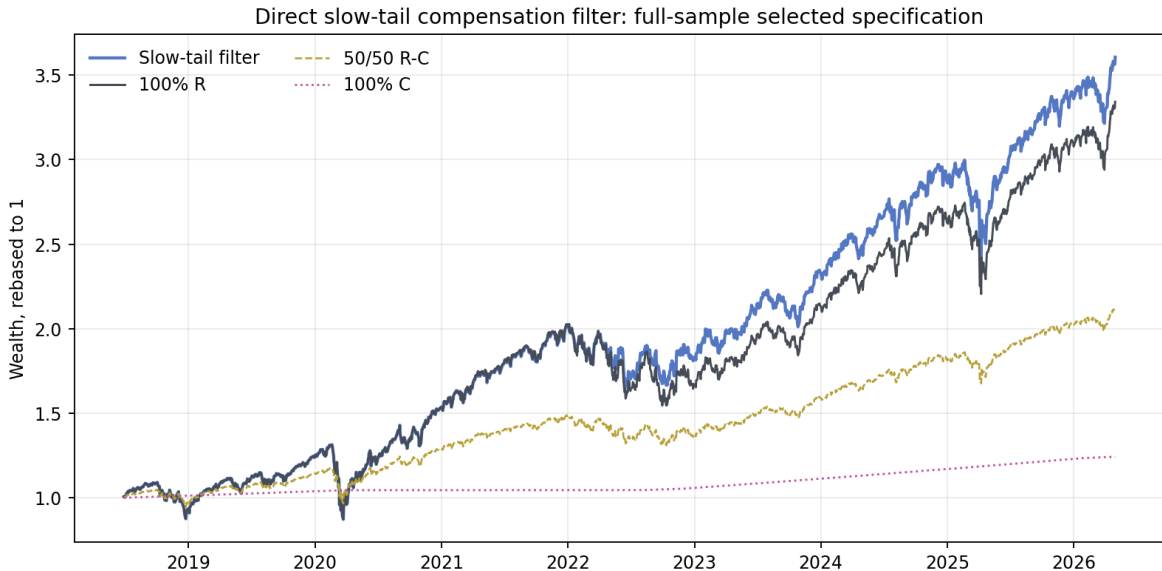


Figure 1: Slow-Tail Selected Full-Sample Equity Curves

The equity curve gives the same interpretation as the parameter and OOS tables. The direct slow-tail rule improves long-run wealth modestly and reduces the 2022 drawdown,

but the full-sample maximum drawdown remains tied to fast crash dynamics (Figure 1). The module is therefore a slow-compensation filter, not a complete drawdown solution.

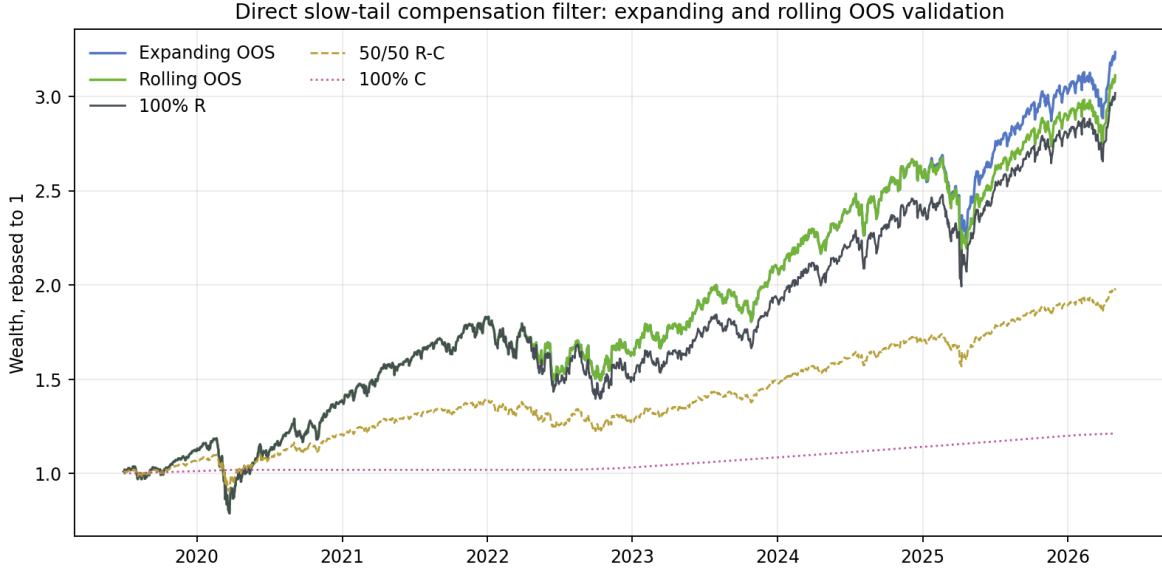


Figure 2: Slow-Tail Walk-Forward Equity Curves

The OOS equity paths emphasize timing rather than only terminal wealth. The expanding and rolling slow-tail paths do not defend every drawdown; their value appears when compensation deterioration is persistent enough to be captured by the direct compensation score (Figure 2).

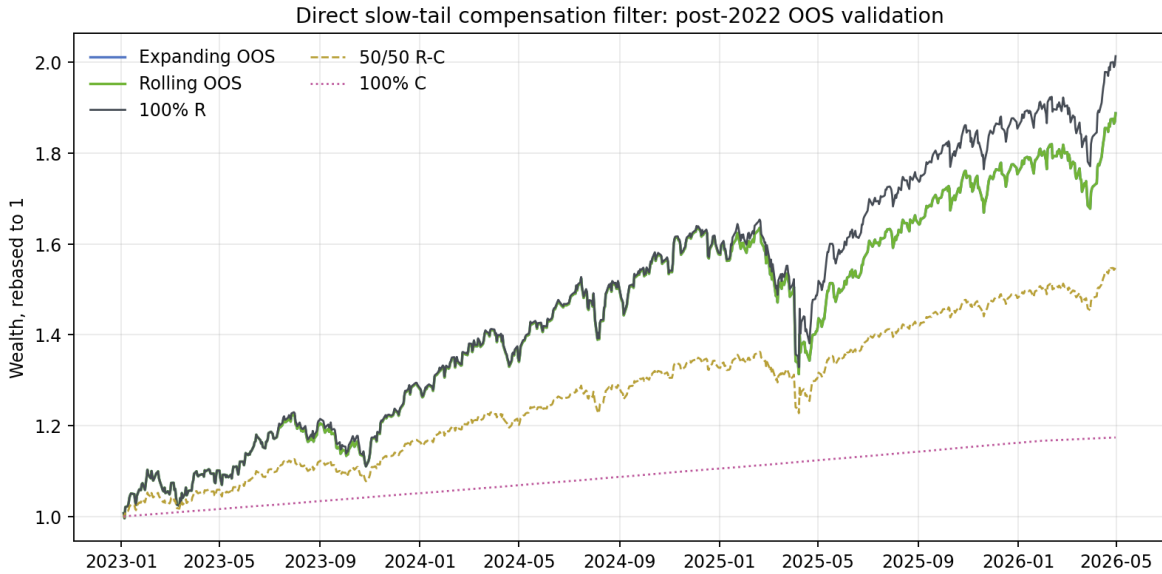


Figure 3: Slow-Tail Post-2022 OOS Equity Curves

The post-2022 window now shows the cost of making the direct slow-tail branch less aggressive. The strategy does not materially improve the recent drawdown path and trails the strong 100% R rebound (Figure 3). This is one of the reasons the combined policy relies on V-shape for fast stress and does not treat slow-tail as a standalone defense.

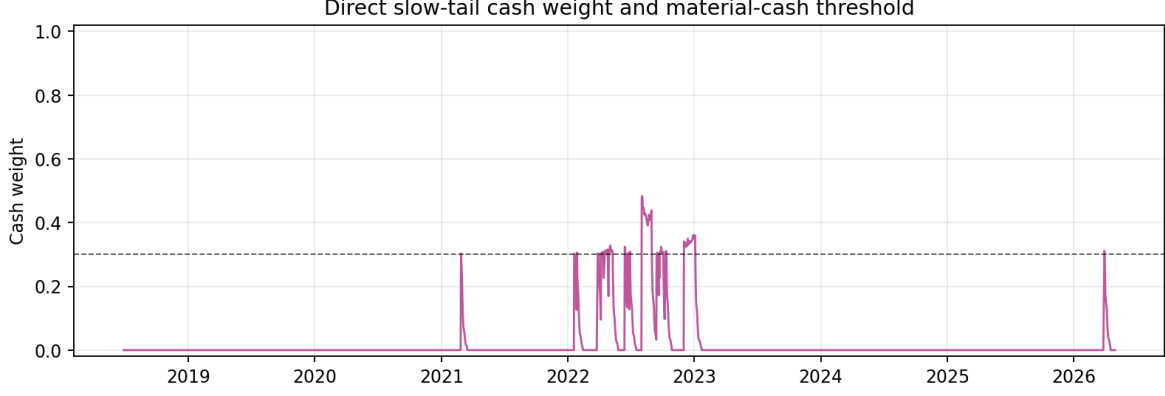


Figure 4: Slow-Tail Selected Cash-Weight Path

The selected slow-tail policy is sparse rather than constantly defensive. Cash exposure clusters in slow compensation-deterioration regimes (Figure 4). Entry occurs once the continuous raw cash weight becomes economically material, and the return toward R is gradual because the exit leg is smoothed.

8 V-Shape Crash-Brake Filter

8.1 Policy Construction

The V-shape filter is designed for fast drawdown events. Unlike slow-tail, it does not estimate a historical analogue mean. It builds a crash-brake score directly from signed market-stress primitives and interaction terms. The signed primitives are

$$VL_t = z_t(VIXPct_{756,t}), \quad VS10_t = z_t(\Delta VIX_{10,t}), \quad (12)$$

$$RR_t = z_t(-\Delta TNX_{21,t}), \quad RL10_t = z_t(-RTrailing_{10,t}), \quad (13)$$

$$CW_t = z_t(\Delta CS_{21,t}), \quad CRO_t = z_t(-HYGSHYRel_{21,t}), \quad (14)$$

$$DD_t = z_t(-RDrawdown_t), \quad VR5_t = z_t(-\Delta VIX_{5,t}), \quad (15)$$

$$CA5_t = z_t(HYGTrailing_{5,t} - SHYTrailing_{5,t}). \quad (16)$$

Here RR is a rate-relief shock, $RL10$ is a 10-day risky-sleeve loss, CRO is credit risk-off, DD is drawdown depth, $VR5$ is VIX relief, and $CA5$ is credit appetite. Candidate products are standardized again before entering the deployable score:

$$BrakeVIXLevel_t = z_t(VL_t), \quad (17)$$

$$BrakeVIXSpike_t = z_t(VS10_t), \quad (18)$$

$$RatePanic_t = 0.5z_t(RR_tRL10_tVL_t) + 0.5z_t(RR_tCRO_t), \quad (19)$$

$$CreditPanic_t = z_t(VS10_tCW_t), \quad (20)$$

$$ReentryOversold_t = z_t(DD_tVL_t), \quad (21)$$

$$ReentryRelief_t = \frac{1}{4}\{z_t(VR5_tCA5_t) + z_t(DD_tVR5_t) \quad (22)$$

$$+ z_t(VL_tVR5_tCA5_t) + z_t(DD_tVR5_tCA5_t)\}. \quad (23)$$

The brake score combines a VIX level/spike core with rate-panic and credit-panic interaction terms:

$$BrakeScore_t = VIXCore_t + \lambda_r RatePanic_t + \lambda_c CreditPanic_t, \quad (24)$$

$$VIXCore_t = \alpha_{vix} BrakeVIXLevel_t + (1 - \alpha_{vix}) BrakeVIXSpike_t. \quad (25)$$

The diagnostic re-entry score is

$$ReentryScore_t = 0.8 ReentryRelief_t + 0.2 ReentryOversold_t, \quad (26)$$

but it is not used as a separate hard trading gate in the final material policy. It is retained to diagnose whether stress is being replaced by rebound conditions. The brake score is expanding-standardized and mapped into a continuous raw cash weight:

$$CashRaw_t = MaxCash \left[\frac{1}{1 + \exp(-BrakeZ_t/\tau)} \right]^\gamma. \quad (27)$$

As in the slow-tail policy, cash is activated only when $CashRaw_t \geq 0.30$. Entry into cash is immediate, while exit back to the risky sleeve is smoothed.

The selected full-sample configuration is `brake_av0.50_lr0.25_lc0.25_cash0.75_tau1.00_pow5.0_mat0.30_ee1.00_ex0.25`. The grid contains 216 configurations over VIX mix, rate-panic weight, credit-panic weight, max cash, brake scale, and cash convexity.

The V-shape tables in this section are the archived standalone module results. They use the V-shape script’s native turnover accounting, which records both the change in risky weight and the offsetting change in cash weight. In the max-cash combination section, the same selected V-shape weight path is recomputed on the shared daily panel using the final one-way R -weight turnover convention. That recomputation is the correct comparison for the combined portfolio.

Table 9: V-Shape Top Configurations by Selection Score

α_{vix}	λ_r	λ_c	Max Cash	τ	Power	CAGR	Score
0.50	0.25	0.25	0.75	1.00	5	17.32%	0.769
0.50	0.50	0.25	1.00	1.00	5	17.07%	0.763
0.50	0.50	0.25	0.75	1.00	5	17.44%	0.758
0.67	0.25	0.25	0.75	1.00	5	17.64%	0.756
0.50	0.25	0.25	0.50	1.00	4	17.67%	0.754
0.50	0.25	0.25	0.50	0.75	5	17.29%	0.752
0.67	0.50	0.25	0.75	1.00	5	17.57%	0.744
0.50	0.25	0.25	0.50	1.00	3	17.27%	0.743

The selected V-shape rules have a clear crash-brake shape. In Table 9, the leading configurations prefer a balanced VIX level/spike core and modest rate/credit panic weights. They also favor high cash convexity, which means cash exposure is not increased linearly for ordinary stress. Instead, the rule waits for an extreme brake score before allocating substantial cash. This is consistent with the intended use case: the module should be quiet most of the time and aggressive only when crash-like conditions appear.

Table 10: V-Shape Event Diagnostics

Event	Strategy Ret.	100% <i>R</i> Ret.	Excess	Avg. Cash	Max Cash	Days
COVID crash	-14.66%	-33.17%	18.50%	58.64%	74.88%	24
April 2025 crash	-10.17%	-14.40%	4.23%	19.33%	74.32%	11
Slow 2022 window	-12.43%	-12.43%	0.00%	0.00%	0.00%	22
COVID recovery	19.75%	27.21%	-7.46%	12.50%	67.54%	28
April 2025 recovery	5.49%	9.90%	-4.41%	41.91%	74.54%	16

Event behavior gives the clearest description of the V-shape module. During the COVID crash, the strategy cuts the drawdown-period loss from -33.17% to -14.66%, with average cash exposure of 58.64% (Table 10). During the April 2025 crash, it also reduces loss, though less dramatically, because the event is shorter and the rebound is fast. The negative recovery rows are the price of this protection: once the rule holds cash during a V-shaped rebound, it can lag the risky sleeve. The zero cash exposure in the slow 2022 window confirms that this module does not solve the 2022-style problem.

Table 11: V-Shape Standalone Walk-Forward Validation

Filter	Mode	CAGR	Max DD	Ann. Turnover	Avg. Cash	100% <i>R</i> CAGR	100% <i>R</i> Max DD
V-shape	Expanding	15.59%	-25.98%	513.18%	3.92%	16.09%	-33.59%
V-shape	Rolling	14.88%	-26.37%	585.82%	4.65%	16.09%	-33.59%

The standalone OOS evidence should therefore be read through a drawdown lens, not as an alpha table. Table 11 shows that both expanding and rolling versions reduce maximum drawdown by roughly seven percentage points relative to 100% *R*, but both trail the risky sleeve on CAGR. That trade-off is acceptable only if the module is combined with a separate return-preserving or slow-tail component, which is exactly why the max-cash combination is tested later.

Table 12: V-Shape Post-2022 Walk-Forward Validation

Filter	Mode	CAGR	Max DD	Ann. Turnover	Avg. Cash	100% <i>R</i> CAGR	100% <i>R</i> Max DD
V-shape	Expanding	10.32%	-25.97%	521.66%	3.15%	12.47%	-23.78%
V-shape	Rolling	8.35%	-26.36%	661.05%	4.67%	12.47%	-23.78%

The post-2022 V-shape result is intentionally weaker. The period contains more slow deterioration than sudden crash onset, so the V-shape brake tends either to stay inactive or to enter too late relative to a slow drawdown (Table 12). This is not a failure of the module’s stated purpose; it is evidence that the module should not be used as the only cash filter.

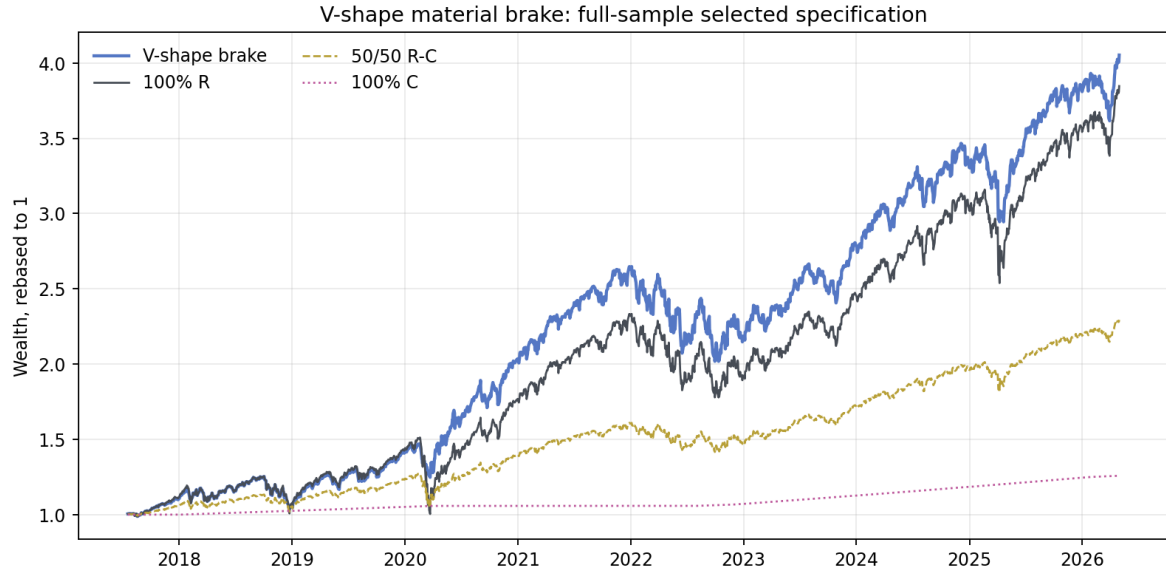


Figure 5: V-Shape Selected Full-Sample Equity Curves

The V-shape module's main trade-off is visible in the selected full-sample path. The strategy lowers fast-crash drawdowns with low average cash exposure, but it can lag during very fast recoveries because cash must be released after stress has already started to normalize (Figure 5). This explains why the module is evaluated as a crash brake rather than as a standalone return enhancer.

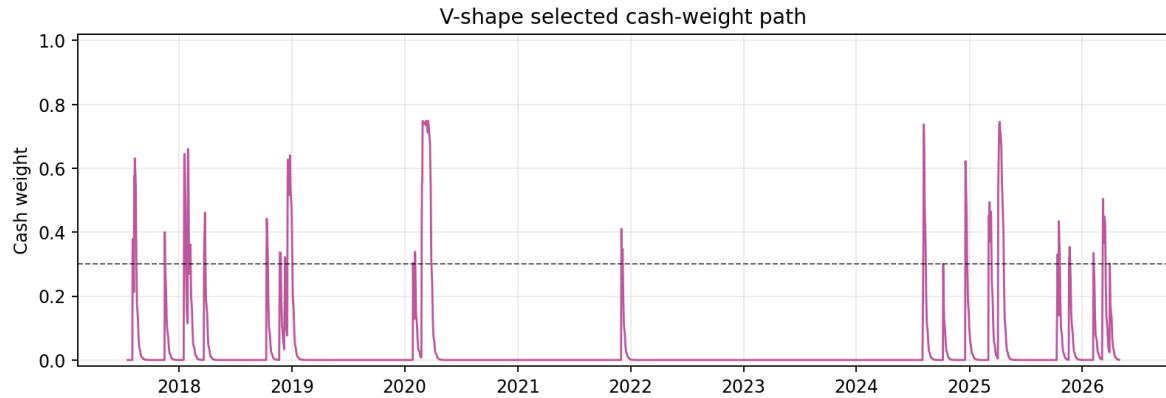


Figure 6: V-Shape Selected Cash-Weight Path

The selected cash-weight path shows how sparse the V-shape brake is in normal markets. Cash exposure remains close to zero most of the time and rises only around short stress episodes, which is consistent with the module's intended role as a temporary crash brake rather than a slow-cycle defensive allocator (Figure 6).

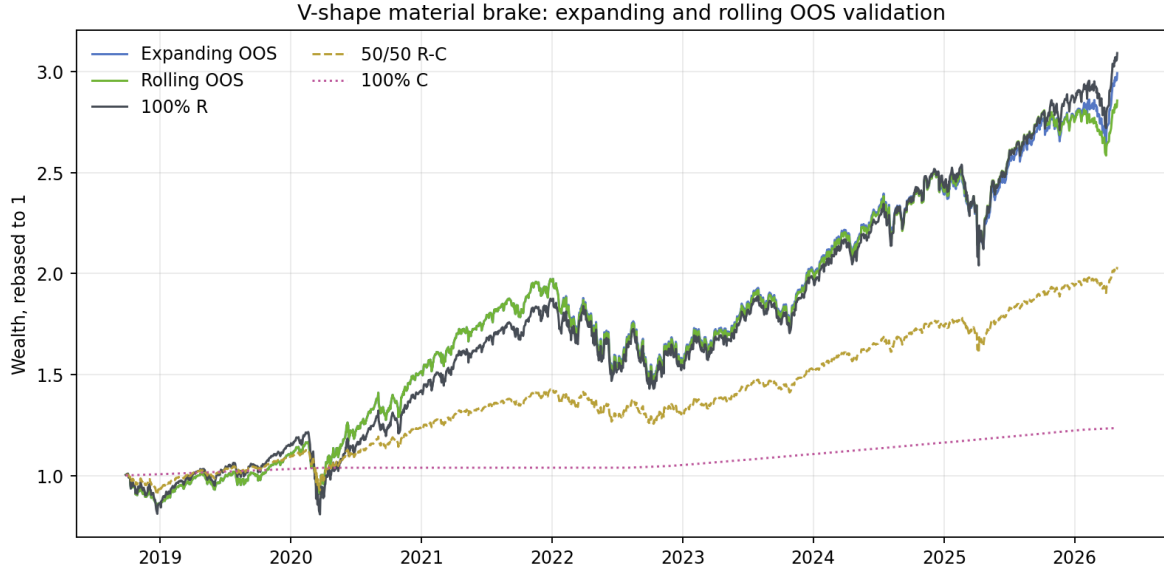


Figure 7: V-Shape Walk-Forward Equity Curves

The OOS paths reinforce the same trade-off. The V-shape strategies reduce drawdown relative to the risky sleeve, but they do not consistently outperform on terminal wealth (Figure 7). This is why the paper later combines V-shape with slow-tail rather than presenting it as a complete cash-allocation policy.

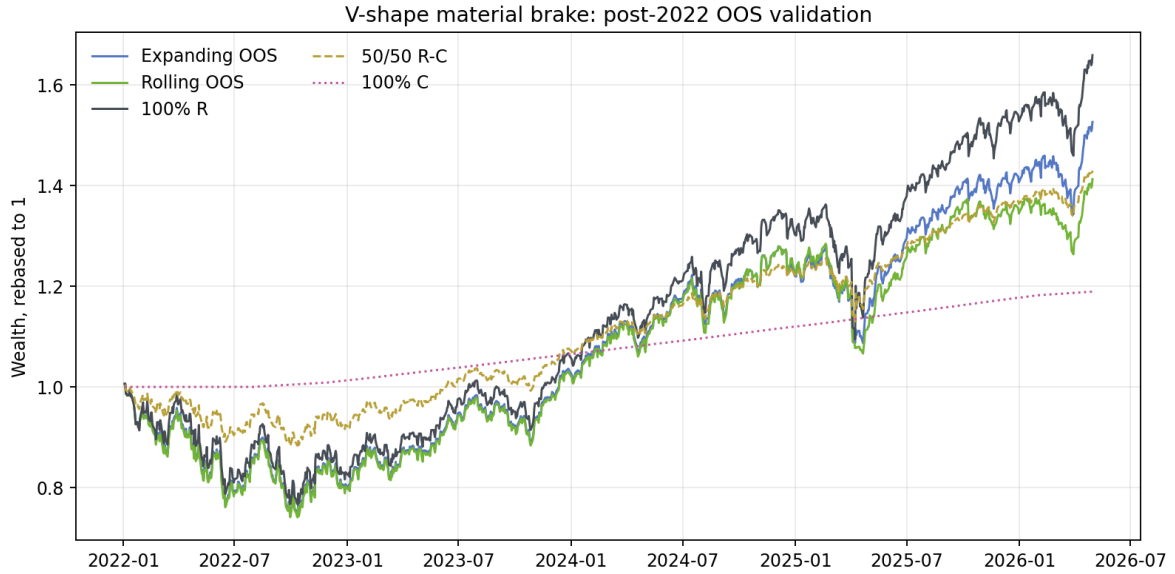


Figure 8: V-Shape Post-2022 OOS Equity Curves

The post-2022 path confirms the limits of the crash-brake design. The window is not one clean V-shaped crash, so the V-shape module can either stay inactive during slow deterioration or create rebound opportunity cost after stress spikes (Figure 8). This visual result is consistent with the event diagnostics in Table 10: the module is useful for fast stress, but it is not a complete cash-allocation policy by itself.

9 Max-Cash Combination

The two filters are combined by a fixed conservative rule:

$$w_t^C = \max(w_t^{C,slow}, w_t^{C,vshape}), \quad w_t^R = 1 - w_t^C. \quad (28)$$

This rule does not select between models. It treats the two filters as independent risk alarms. If either module demands a larger cash allocation, the final portfolio uses that larger cash weight.

Table 13: Max-Cash Selected-Weight Combination, Common Window

Method	CAGR	Max DD	Ann. Turnover	Avg. Cash	Max Cash	Final Wealth
Max-cash	18.83%	-18.05%	355.82%	5.56%	74.88%	4.54
Slow-tail only	17.63%	-33.59%	96.68%	1.96%	48.33%	4.15
V-shape only	17.63%	-23.77%	266.20%	3.65%	74.88%	4.15
100% R	16.62%	-33.59%	0.00%	0.00%	0.00%	3.85

The combination is economically different from either component alone. Table 13 recomputes slow-tail, V-shape, max-cash, and the benchmarks on the same common daily panel with the final one-way turnover convention. This is why the V-shape-only CAGR and turnover differ from the standalone V-shape table above even though the selected weight path is the same. Under this common convention, the redesigned slow-tail module improves return but does not reduce the main maximum drawdown; V-shape reduces drawdown but is not a 2022 slow-tail solution. The max-cash rule keeps the stronger defensive action each day, so its maximum drawdown is lower than either component and its CAGR remains above 100% R . The stricter OOS-weight combination is reported next.

Table 14: Key Event Diagnostics for Max-Cash Combination

Event	Policy Ret.	100% R Ret.	Policy DD	100% R DD	Avg. Cash
COVID crash, 2020-02-19–2020-03-23	-14.58%	-33.17%	-15.11%	-33.59%	58.64%
Calendar 2022	-8.95%	-17.06%	-18.05%	-23.78%	15.60%
Slow 2022, 2022-08-25–2022-09-26	-8.78%	-12.43%	-9.59%	-13.75%	24.38%
April 2025 crash, 2025-03-25–2025-04-08	-10.10%	-14.40%	-10.11%	-14.41%	19.33%
April 2025 crash plus recovery	-4.66%	-4.20%	-10.11%	-14.41%	31.11%

The event decomposition clarifies why a simple max-cash rule is enough. In Table 14, the COVID crash benefit comes almost entirely from the fixed V-shape module, while the 2022 benefit comes from the redesigned slow-tail branch. The current direct slow-tail version no longer eliminates the 2022 drawdown as aggressively as the previous analogue version, but it still improves both the full calendar 2022 and the August–September slow-tail window. The combined rule does not need to identify the event type in advance; it simply takes the larger cash weight from whichever module is active.

Table 15: Max-Cash Rule Applied to Sub-Strategy OOS Weights

OOS Window	Mode	Method	CAGR	Max DD	Ann. Turnover	Avg. Cash
main-oos	Expanding	Max-cash	19.35%	-22.05%	305.09%	5.22%
main-oos	Expanding	100% <i>R</i>	17.59%	-33.59%	0.00%	0.00%
main-oos	Rolling	Max-cash	18.50%	-22.05%	351.81%	6.15%
main-oos	Rolling	100% <i>R</i>	17.59%	-33.59%	0.00%	0.00%
post2022-oos	Expanding	Max-cash	20.76%	-15.53%	397.21%	4.56%
post2022-oos	Expanding	100% <i>R</i>	23.58%	-19.66%	0.00%	0.00%
post2022-oos	Rolling	Max-cash	18.91%	-17.26%	470.06%	6.24%
post2022-oos	Rolling	100% <i>R</i>	23.58%	-19.66%	0.00%	0.00%

The OOS-weight combination is the more conservative comparison. Here the component weights are not the selected full-sample paths; they are generated by each component’s own expanding or rolling walk-forward selection (Table 15). The main OOS window improves both CAGR and drawdown for both expanding and rolling versions. In the post-2022 window, both expanding and rolling lower drawdown but sacrifice CAGR relative to 100% *R*. This is the key cost of the cash overlay during a strong risky-sleeve rebound.

Table 16: Component Trigger Overlap, 1% Cash-Weight Threshold

Bucket	Days	Sample Share	Avg. Combined Cash
Slow-tail only	189	8.56%	22.07%
V-shape only	347	15.72%	22.74%
Both active	13	0.59%	11.46%
Combined active	549	24.86%	22.24%

Redundancy is low despite the simple combination rule. At the reported active-state threshold, Table 16 shows that both modules are active together on only 13 trading days, or 0.59% of the common window. The combined active share of 24.86% is therefore not produced by two copies of the same signal. The modules fire in different market states, which supports the max-cash combination rule.

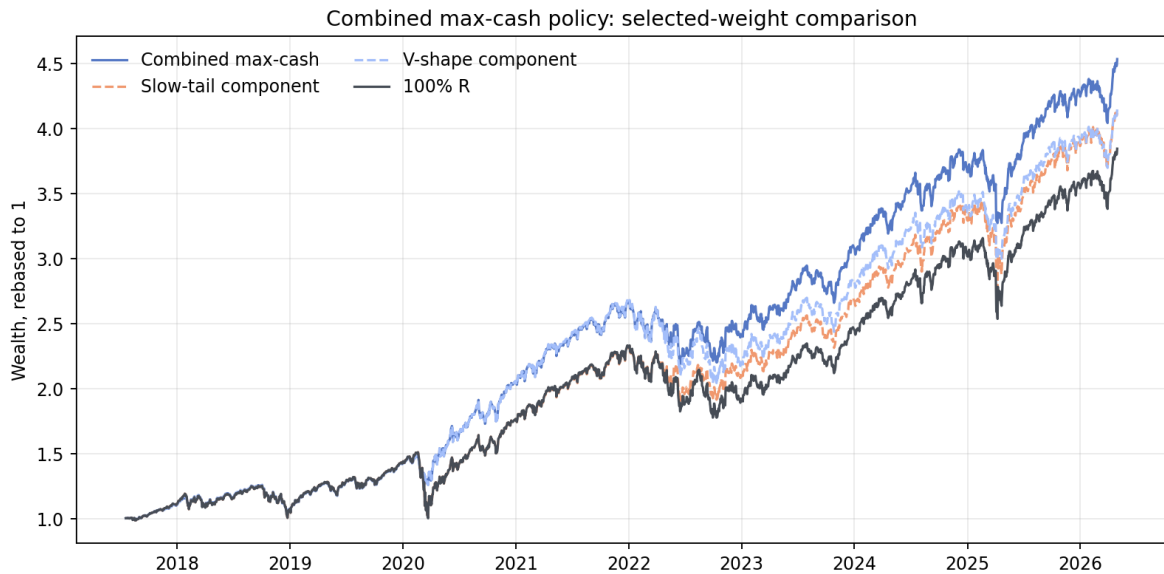


Figure 9: Selected-Weight Max-Cash Combination Equity Curves

The selected-weight equity paths show how the upper-envelope rule works in practice. The combined line is not an average of the two filters; it follows the larger cash demand each day (Figure 9). This is why the combined path can keep the slow-tail 2022 benefit and the V-shape fast-crash benefit in the same portfolio.

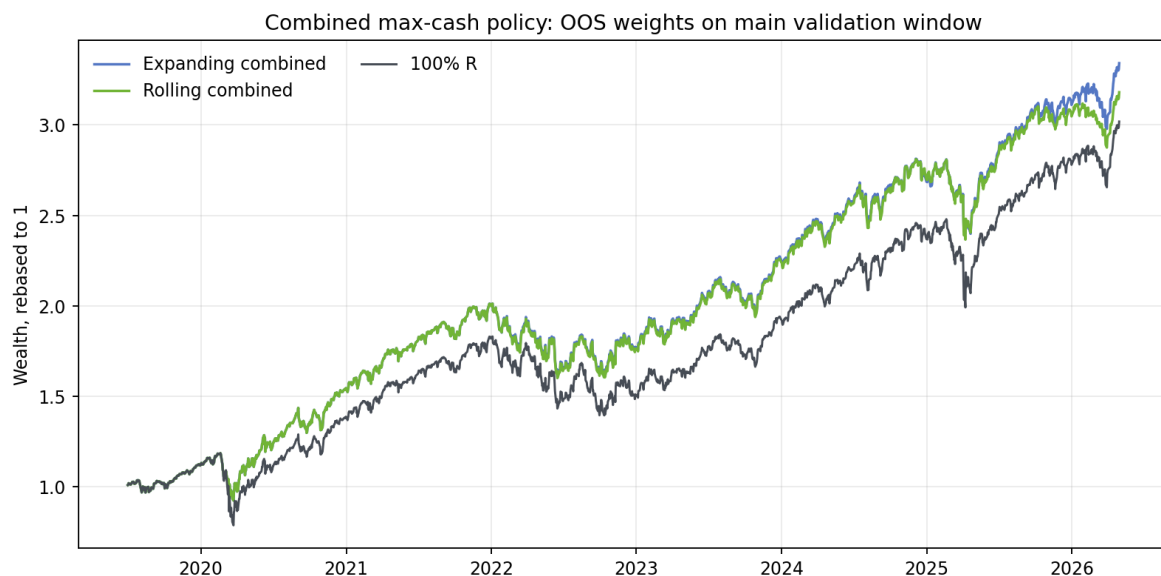


Figure 10: Main OOS Max-Cash Combination Equity Curves

The stricter OOS-weight construction is shown separately. Each component first selects parameters using only past data, and the max-cash rule is then applied to those OOS weights (Figure 10). This checks whether the modular combination survives after removing the selected full-sample component paths.

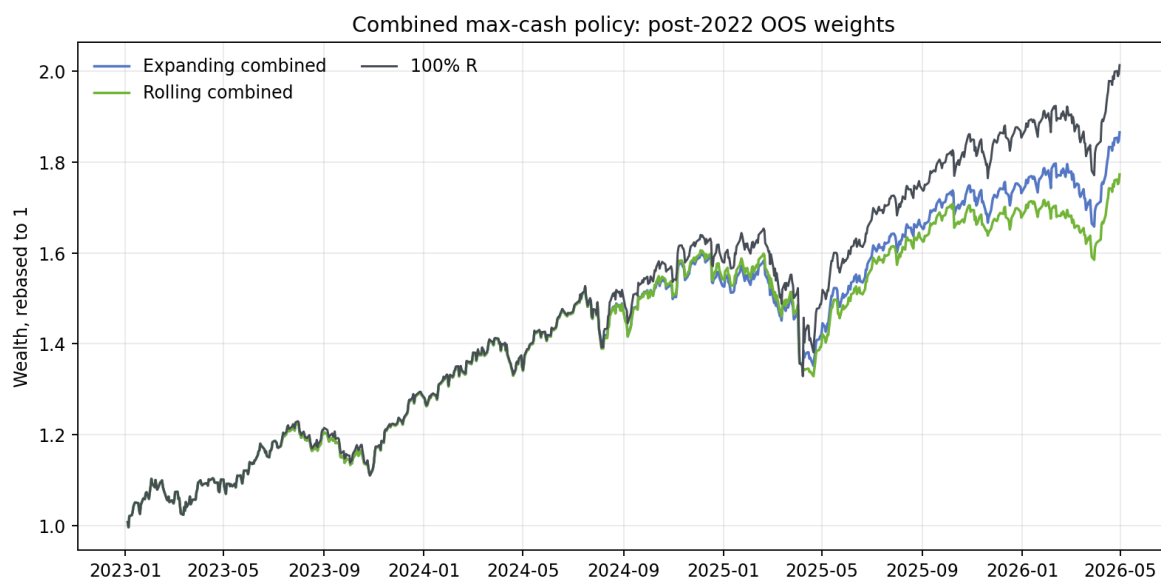


Figure 11: Post-2022 OOS Max-Cash Combination Equity Curves

The post-2022 OOS-weight comparison separates drawdown control from return capture. Both expanding and rolling versions reduce drawdown versus 100% R , but both trail

the strong risky-sleeve rebound in terminal wealth (Figure 11). This visual distinction matches the numerical comparison in Table 15.

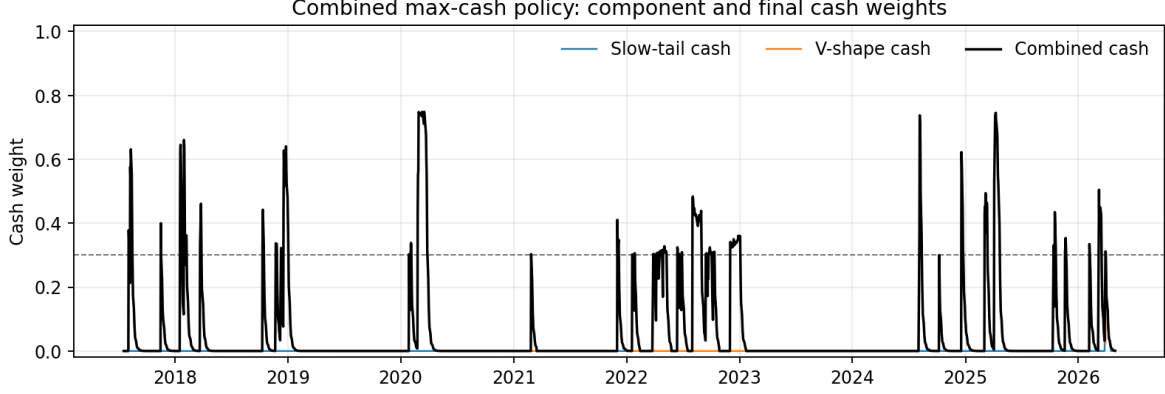


Figure 12: Selected-Weight Component and Max-Cash Allocations

The cash-weight path makes the mechanics behind the combined equity curve explicit. The combined cash weight is the upper envelope of the slow-tail and V-shape cash weights (Figure 12). This preserves specialized alarms: slow-tail can dominate during persistent weak compensation, while V-shape can dominate during fast stress events.

10 Rolling Start-Date Sensitivity

The companion style-timing study reports start-date sensitivity because a single OOS start can hide dependence on block boundaries [Xiong(2026)]. The same diagnostic is repeated here. For each requested start date, the two component policies reselect the full score and position-parameter vector in each 63-trading-day OOS block. The rolling version uses the most recent 756 trading days as the training window. The max-cash combination is then rebuilt from those two component OOS weight paths.

The rolling result is economically useful but not uniform. Because the direct slow-tail module requires a 252-trading-day warmup inside each requested-start experiment, the first effective OOS date is about one year after the requested start for the combined path. For requested starts through 2019, rolling max-cash improves both CAGR and maximum drawdown relative to 100% R . For the 2020 and 2022 requested starts, it still reduces drawdown but gives up CAGR, which is consistent with a short training window that becomes more defensive after recent stress.

Table 17: Rolling Start-Date Sensitivity: Start-to-End CAGR and Drawdown

Requested Start	Max-Cash CAGR	Sharpe	Max DD	CAGR Diff	MDD Improvement	100% R CAGR	100% R Max DD
2018-06-28	18.50%	1.09	-22.05%	0.90%	11.53%	17.59%	-33.59%
2018-07-30	18.61%	1.09	-22.05%	1.35%	11.53%	17.27%	-33.59%
2018-08-29	19.50%	1.14	-22.05%	1.37%	11.53%	18.13%	-33.59%
2018-10-29	19.36%	1.12	-22.05%	1.63%	11.53%	17.74%	-33.59%
2019-01-02	17.62%	1.04	-22.05%	0.63%	11.53%	16.99%	-33.59%
2020-01-02	15.20%	1.01	-19.18%	-0.64%	4.60%	15.84%	-23.78%
2022-01-03	18.91%	1.40	-17.26%	-4.67%	2.41%	23.58%	-19.66%

Across requested starts, the rolling version is robust for drawdown control but not for

return dominance. Table 17 shows positive CAGR improvement for 2018–2019 requested starts, while the 2020 and 2022 requested starts lag 100% R after still reducing drawdown. This pattern means the rolling window is useful as a recent-regime adapter, but it can become too anchored to recent stress or miss part of a strong rebound.

Table 18: Rolling Common-Window Sensitivity, 2022-01-03 to 2026-04-30

Requested Start	Max-Cash CAGR	Max DD	100% R CAGR	50/50 $R - C$ CAGR	100% C CAGR	CAGR Diff	MDD Improvement
2018-06-28	11.38%	-20.42%	12.47%	8.65%	4.11%	-1.10%	3.36%
2018-07-30	12.34%	-20.83%	12.47%	8.65%	4.11%	-0.14%	2.95%
2018-08-29	12.35%	-19.70%	12.47%	8.65%	4.11%	-0.13%	4.08%
2018-10-29	13.23%	-18.33%	12.47%	8.65%	4.11%	0.76%	5.45%
2019-01-02	12.04%	-19.18%	12.47%	8.65%	4.11%	-0.43%	4.60%
2020-01-02	12.05%	-19.18%	12.47%	8.65%	4.11%	-0.43%	4.60%
2022-01-03	18.91%	-17.26%	23.58%	14.23%	4.97%	-4.67%	2.41%

Putting all rolling variants on the same post-2022 evaluation window isolates the effect of the requested start from the effect of different calendar samples. The common-window comparison in Table 18 shows the cost of the current direct slow-tail design most clearly: drawdown improvement is generally positive, but CAGR superiority appears only for one requested-start variant and is lost when the risky sleeve rebounds strongly.

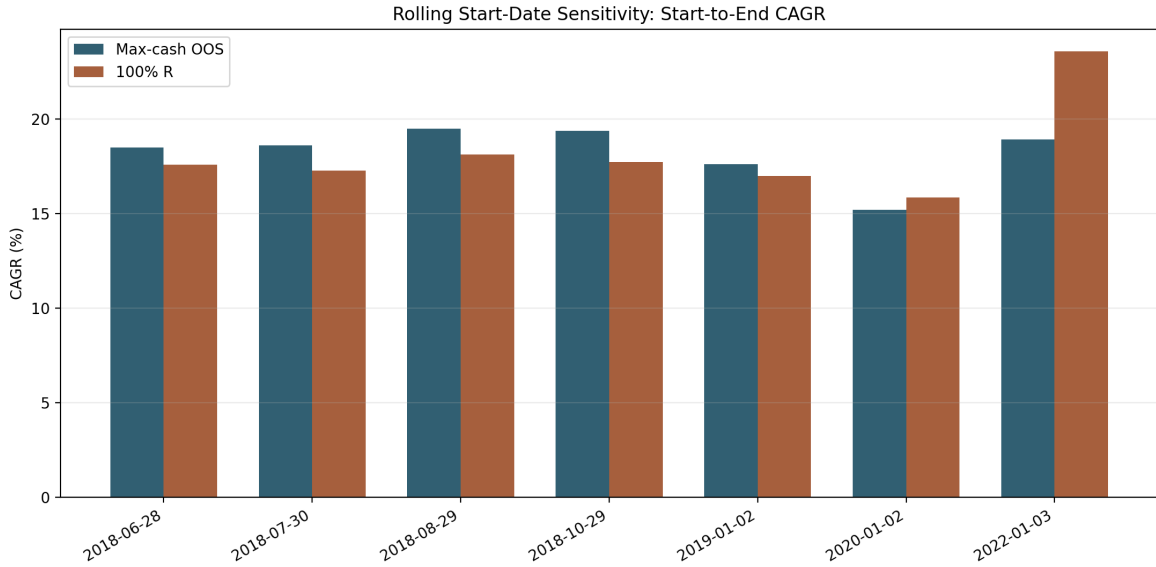


Figure 13: Rolling Start-Date Sensitivity: Start-to-End CAGR

The rolling start-to-end CAGR comparison is visually asymmetric. The rolling max-cash rule beats 100% R for requested starts through 2019, while later requested starts are weaker (Figure 13). This highlights the same caution as Table 17: rolling selection adapts quickly but may become over-defensive after a stress-heavy training window.

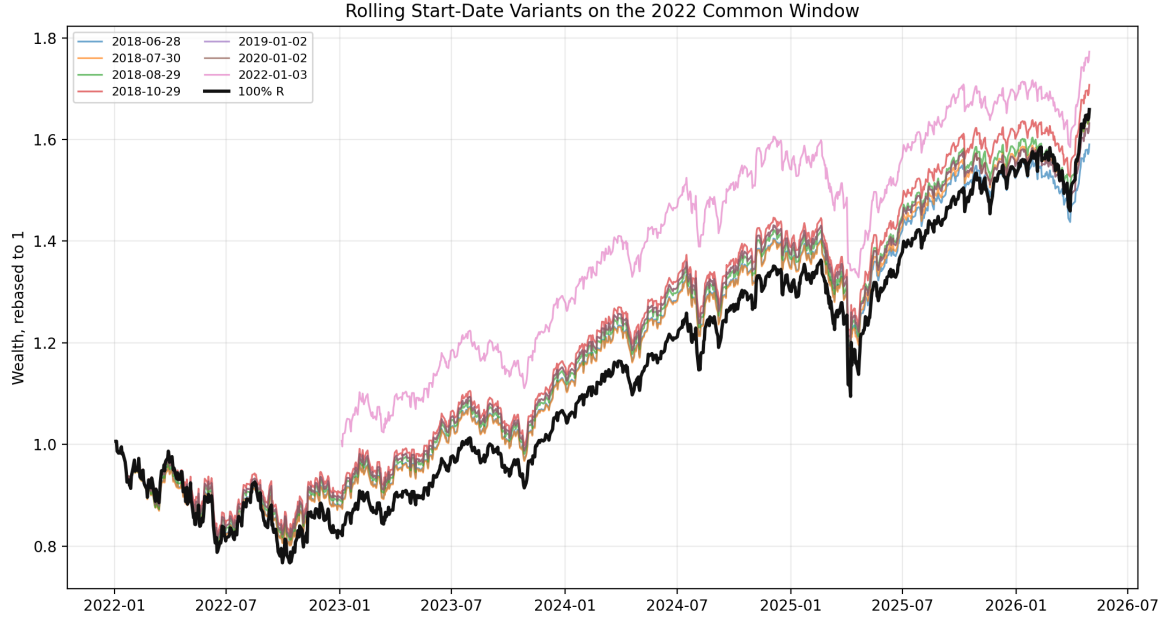


Figure 14: Rolling Start-Date Variants on the Post-2022 Common Window

Rebasing all rolling requested-start variants on the same post-2022 window removes sample-length differences. In Figure 14, most variants retain some drawdown control, but several lag the 100% R rebound. The common-window view therefore separates defensive usefulness from return dominance.

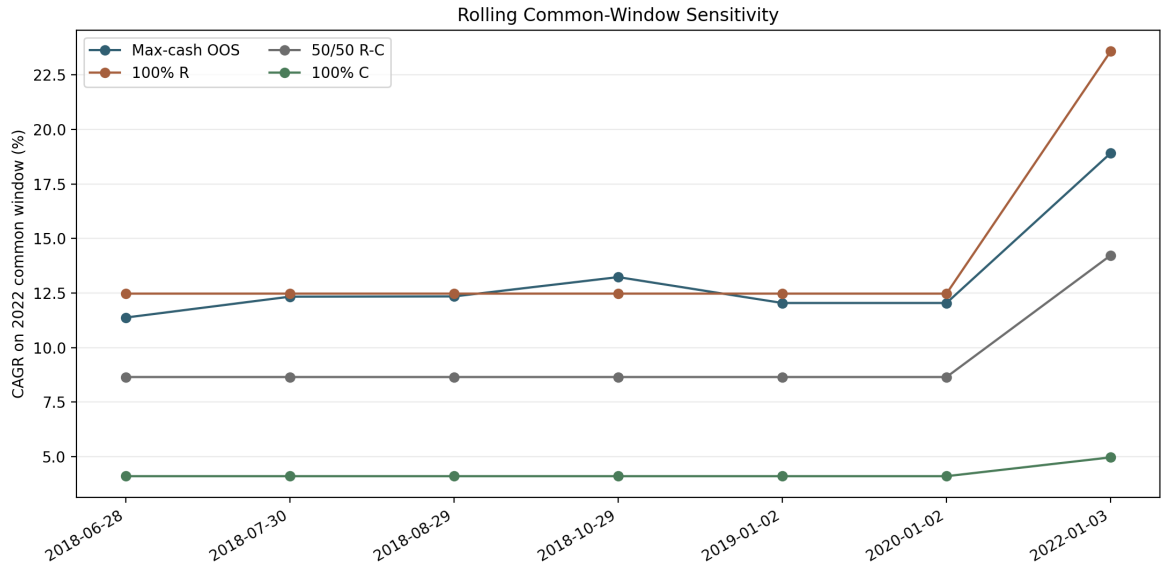


Figure 15: Rolling Common-Window CAGR Sensitivity

The same common-window rolling test can be summarized in CAGR terms. Figure 15 shows that the current rolling specifications often fall below 100% R on post-2022 CAGR even when they reduce drawdown. This is a risk-management trade-off rather than a standalone alpha result.

11 Expanding Start-Date Sensitivity

The expanding version uses all available past data before each OOS block. This is the cleaner validation of a slowly accumulating empirical rule: every later block has a larger training sample, and the choice of requested start mainly changes the early OOS blocks and block alignment. In the current direct slow-tail version, expanding selection is stronger than rolling on the main window, but it no longer improves CAGR for every requested start. Its more stable result is drawdown reduction.

This does not remove model-selection risk, because the variable families and policy classes were constructed before the walk-forward exercise. It does show, however, that once the two module classes are fixed, the expanding parameter-selection process is not fragile to reasonable OOS start-date choices.

Table 19: Expanding Start-Date Sensitivity: Start-to-End CAGR and Drawdown

Requested Start	Max-Cash CAGR	Sharpe	Max DD	CAGR Diff	MDD Improvement	100% <i>R</i> CAGR	100% <i>R</i> Max DD
2018-06-28	19.35%	1.13	-22.05%	1.76%	11.53%	17.59%	-33.59%
2018-07-30	18.72%	1.10	-22.05%	1.45%	11.53%	17.27%	-33.59%
2018-08-29	19.94%	1.16	-22.05%	1.81%	11.53%	18.13%	-33.59%
2018-10-29	19.29%	1.12	-22.05%	1.56%	11.53%	17.74%	-33.59%
2019-01-02	18.13%	1.07	-22.05%	1.14%	11.53%	16.99%	-33.59%
2020-01-02	15.79%	1.05	-19.98%	-0.05%	3.79%	15.84%	-23.78%
2022-01-03	20.76%	1.50	-15.53%	-2.82%	4.13%	23.58%	-19.66%

The expanding start-date result is stronger than the rolling counterpart on early requested starts. In Table 19, requested starts through 2019 improve both CAGR and maximum drawdown relative to 100% *R*. The 2020 and 2022 requested starts still reduce drawdown but trail on CAGR, which is consistent with the post-stress rebound cost documented in the component-level OOS tests.

Table 20: Expanding Common-Window Sensitivity, 2022-01-03 to 2026-04-30

Requested Start	Max-Cash CAGR	Max DD	100% <i>R</i> CAGR	50/50 <i>R</i> - <i>C</i> CAGR	100% <i>C</i> CAGR	CAGR Diff	MDD Improvement
2018-06-28	12.65%	-20.03%	12.47%	8.65%	4.11%	0.18%	3.75%
2018-07-30	12.48%	-20.31%	12.47%	8.65%	4.11%	0.01%	3.47%
2018-08-29	12.97%	-19.28%	12.47%	8.65%	4.11%	0.50%	4.49%
2018-10-29	13.12%	-18.36%	12.47%	8.65%	4.11%	0.64%	5.41%
2019-01-02	12.74%	-19.98%	12.47%	8.65%	4.11%	0.27%	3.79%
2020-01-02	12.74%	-19.98%	12.47%	8.65%	4.11%	0.27%	3.79%
2022-01-03	20.76%	-15.53%	23.58%	14.23%	4.97%	-2.82%	4.13%

On the post-2022 common window, expanding selection is less sensitive to the requested start than rolling selection, but the result is still not a pure return improvement. Table 20 reports modest CAGR gains for pre-2022 requested starts and a material CAGR drag for the 2022 requested start, while every row improves maximum drawdown. This is the cleanest validation layer for the current defensive interpretation: the overlay buys drawdown reduction at the cost of some participation in the strongest rebound window.

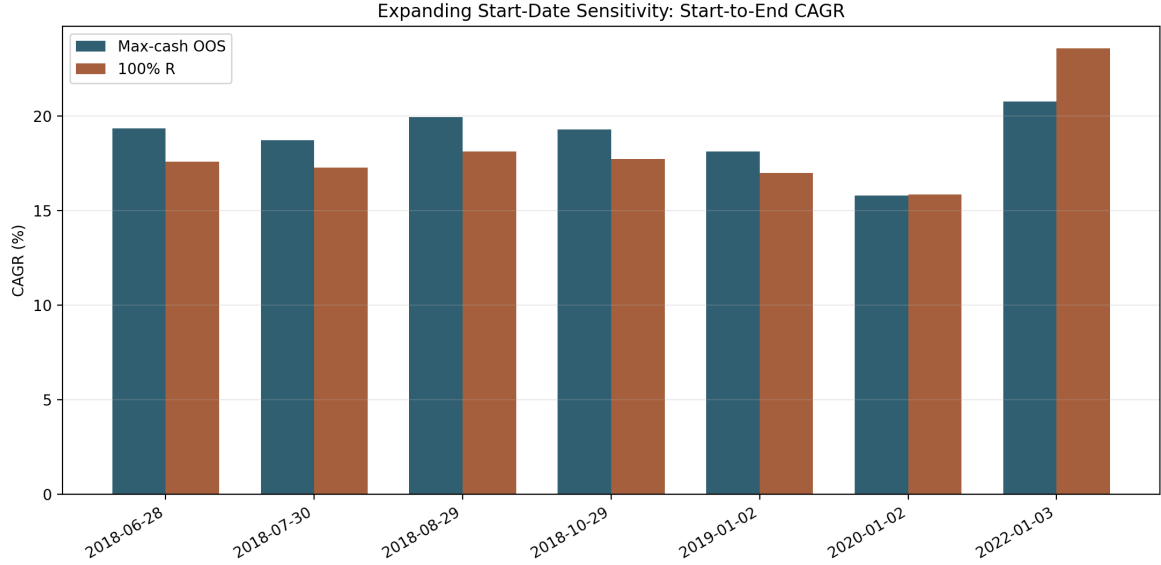


Figure 16: Expanding Start-Date Sensitivity: Start-to-End CAGR

The stronger early-start result is also visible graphically. Expanding max-cash outperforms 100% R for requested starts before 2020, while later starts show the rebound-cost limitation (Figure 16). The figure therefore supports a narrower claim than the earlier version: expanding selection is more stable than rolling, but it is still a defensive overlay.

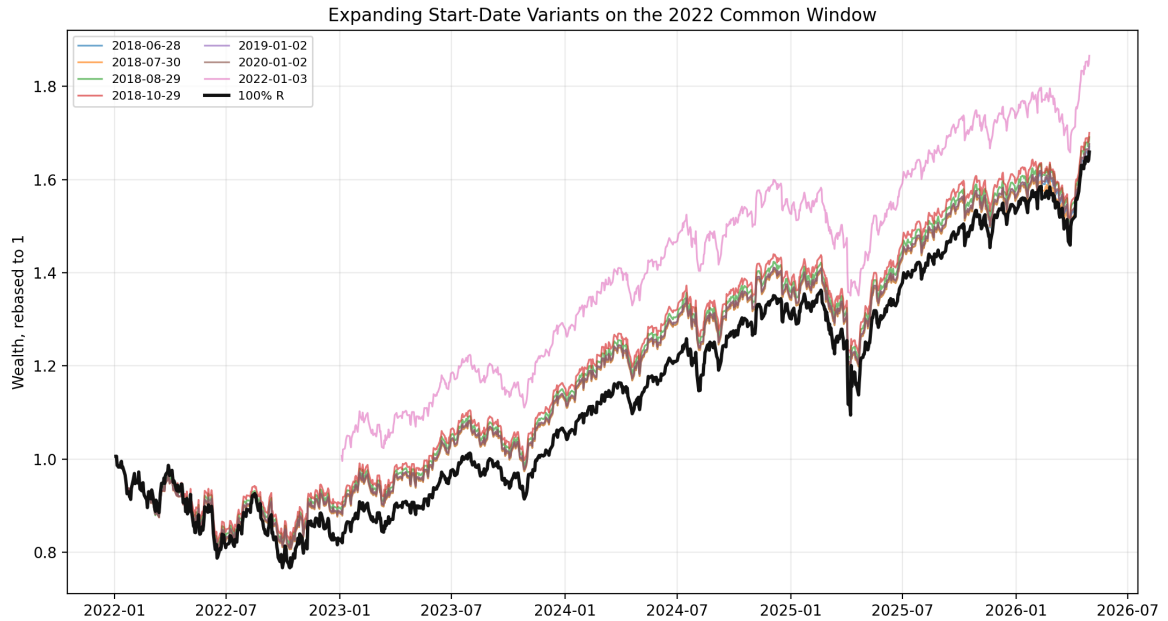


Figure 17: Expanding Start-Date Variants on the Post-2022 Common Window

On the common post-2022 window, the expanding variants cluster more tightly than the rolling variants. This pattern in Figure 17 implies that the expanding result is not driven by one arbitrary requested start, but it also shows that the current direct slow-tail branch cannot dominate a strong risky-sleeve rebound on CAGR.

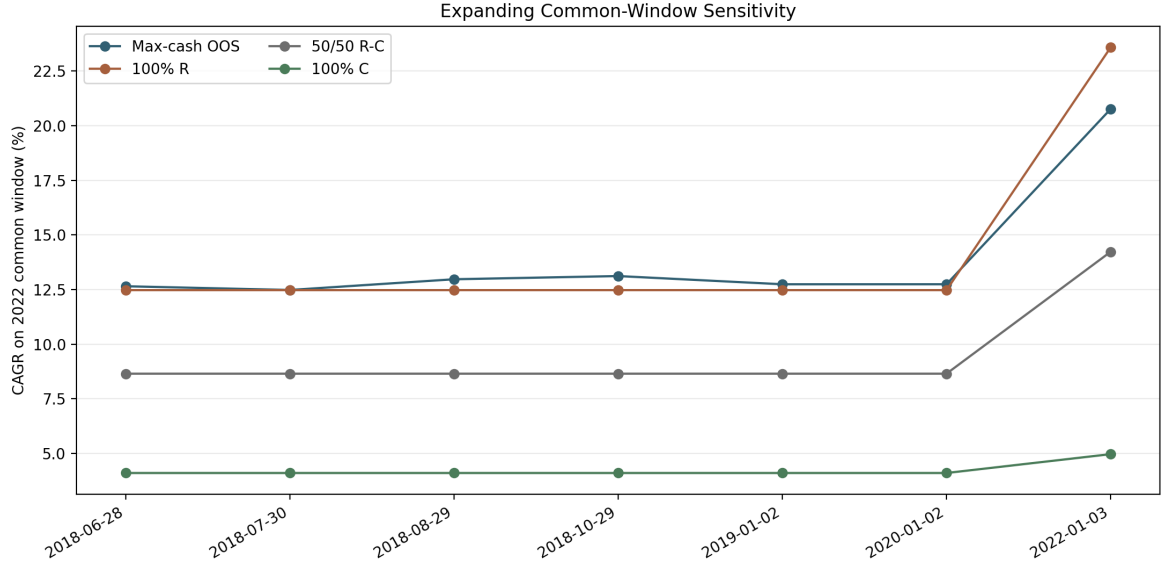


Figure 18: Expanding Common-Window CAGR Sensitivity

The common-window CAGR comparison gives the same conclusion in a more compact form. Expanding selection is close to or above 100% R for pre-2022 requested starts, but the 2022 requested start trails the rebound (Figure 18). This supports using expanding selection as the cleaner validation baseline for the cash-overlay setting, while still treating the overlay as a defensive tool rather than a guaranteed return enhancer.

12 Economic Interpretation and Limitations

The slow-tail filter and V-shape filter cover different risk shapes. The slow-tail filter is most useful when the risky sleeve is not sufficiently compensated over cash, as in 2022. The V-shape filter is most useful when fast stress creates a large drawdown risk, as in 2020 and April 2025. Their active cash-weight overlap is small, so the combined rule is not merely duplicating the same signal.

The economic value is mainly drawdown control with selective return improvement. The full-sample max-cash combination improves both CAGR and maximum drawdown. In walk-forward OOS tests, the main-window expanding and rolling combinations also improve both CAGR and maximum drawdown. Post-2022, however, both expanding and rolling lower drawdown but sacrifice CAGR. This is an important limitation of the current cash-overlay design: it can protect capital in stress, but it can trail a strong risky-sleeve rebound when cash exposure lingers.

The main limitation is data mining. Variables, interactions, and policy structures were developed on the available sample. The walk-forward tests re-select parameters from past data, but the current study does not yet perform real-time expanding re-screening of variables and interactions. It also does not yet report White's Reality Check, Hansen's SPA test, block-bootstrap inference, or the Deflated Sharpe Ratio. The evidence should therefore be read as a structured empirical strategy study, not as a final multiple-testing-adjusted anomaly claim.

13 Conclusion

This paper rewrites the cash-overlay experiment around a clean empirical object: a static equal-weight growth–defensive risky sleeve versus interest-bearing cash. The main methodological update is the direct slow-tail compensation module. It improves the 2022-style low-compensation regime less aggressively than the previous analogue construction, but with a simpler continuous score. The fixed V-shape module continues to reduce fast-crash losses in 2020 and 2025. The max-cash combination improves full-sample CAGR from 16.62% to 18.83% and improves maximum drawdown from -33.59% to -18.05%. In the stricter OOS-weight combination, the main expanding and rolling tests also improve both CAGR and drawdown relative to 100% R .

The results support modular continuous cash overlays as economically meaningful risk-management tools. The next research step is to make variable selection itself real-time, add formal multiple-testing correction, and test the same design on longer or cross-asset histories.

A Screening Tables and Robustness Details

A.1 Screening Protocol

The screening layer is diagnostic rather than a direct trading rule. Each candidate main effect or interaction is evaluated against future $R - C$ returns at 21, 63, and 126 trading-day horizons. The 63-day horizon is the main economic horizon because it is long enough to capture a regime response but short enough to be relevant for tactical allocation. Since future labels overlap at daily frequency, ordinary OLS standard errors are not used as the primary inference object. The reported t -statistics use Newey–West/HAC standard errors.

For a candidate variable x_t , the basic diagnostic regression is

$$R_{t,t+h}^{R-C} = \alpha_h + \beta_h x_t + \epsilon_{t,h}, \quad h \in \{21, 63, 126\}. \quad (29)$$

The coefficient sign, HAC t -value, non-overlapping 63-day coefficient, and multi-horizon direction are all reviewed. Interactions are included only when they have an economic interpretation. For example, a credit-panic term must combine market stress with credit deterioration; a low-compensation term must combine cash attractiveness, risky-sleeve strength, or low-volatility conditions that plausibly indicate limited incremental reward for staying fully in R .

The gate is intentionally weaker than a publication-grade multiple-testing claim. A variable can be retained as a component even when one diagnostic is mixed if the economic role is clear and the component improves portfolio behavior inside a continuous score. Conversely, a statistically strong variable is not automatically retained if it duplicates another signal, has poor coverage, or maps poorly into an implementable cash weight. This follows the workflow of the companion style-timing study [Xiong(2026)]: statistical screens narrow the candidate set, while walk-forward policy performance determines whether the candidate survives as a tradable module.

A.2 Slow-Tail Component Sets

The current slow-tail module is not built around historical analogues. It uses a fixed set of 11 direct continuous components, grouped into cash-rising pressure, compressed risky-sleeve compensation, a direct rate headwind, and rate-path stress. The notation follows the construction in the main text: CH is high cash yield, CR is cash yield rising, RH is rate headwind, DD is drawdown depth, VS is a VIX spike, CRO is credit risk-off, RS is risky-sleeve strength, RU is risky-sleeve underwater, LV is low VIX, and $LVOL$ is low risky-sleeve volatility. Every component listed here is continuous; after products are formed, they are expanding-standardized before entering the direct slow-tail score.

Table 21: Direct Slow-Tail Components Used in the Policy Grid

Block	Component Formula	Economic Meaning
cash rising	CR_t	Cash yield is rising, so the required compensation for holding R over C is higher.
compressed compensation	$RS_t \times LVOL_t$	Risky sleeve has been strong in a low-volatility state; forward compensation may be compressed.
compressed compensation	$RS_t \times LVOL_t \times LV_t$	Strong R , low realized volatility, and low VIX jointly describe complacent high-risk exposure.
compressed compensation	$RS_t \times LV_t$	Risky-sleeve strength when volatility risk is priced cheaply.
compressed compensation	$CH_t \times RS_t \times LV_t$	High cash yield plus strong risky sleeve plus low VIX; cash becomes competitive while risk appetite is still calm.
compressed compensation	$CH_t \times RS_t \times LVOL_t$	High cash yield plus strong risky sleeve plus low realized volatility.
rate headwind	RH_t	Rising 10-year yields as a direct rate headwind.
rate path	$RH_t \times VS_t$	Rate headwind occurring with a VIX spike.
rate path	$RH_t \times CRO_t$	Rate headwind occurring with credit risk-off.
rate path	$RH_t \times RU_t$	Rate headwind while the risky sleeve is already underwater.
rate path	$DD_t \times RH_t \times CRO_t$	Drawdown, rate headwind, and credit risk-off occurring together.

The broader feature panel still contains legacy analogue columns for reproducibility, but they are not used by the current deployable slow-tail policy. The current score uses the component formulas listed in Table 21, the grid weights α_{comp} , λ_{rate} , and λ_{path} , and the continuous cash mapping described in the main text.

A.3 V-Shape Brake and Re-Entry Candidate Families

The V-shape module is designed for fast drawdowns and fast recoveries. Its final material version uses the brake score as the trading score and keeps re-entry terms as diagnostics. The reason is practical: once the brake-derived raw cash weight falls below the 30% material threshold, the policy begins returning to R through the smoothing rule. A second hard re-entry trigger would add an additional threshold, which would conflict with the goal of keeping the final policy mostly continuous. In this table, $EZ(x)$ denotes the second expanding standardization applied after the primitive or product has been constructed.

Table 22: V-Shape Candidate Families

Family	Component Formula	Economic Meaning
VIX brake core	$BrakeVIXLevel_t = EZ(VL_t)$	High VIX level, where $VL_t = z(VIXPct_{756,t})$.
VIX brake core	$BrakeVIXSpike_t = EZ(VS10_t)$	Fast VIX spike, where $VS10_t = z(\Delta VIX_{10,t})$.
Rate panic	$EZ(RR_t \times RL10_t \times VL_t)$	Rate relief, risky-sleeve 10-day loss, and high VIX occurring together.
Rate panic	$EZ(RR_t \times CRO_t)$	Rate relief occurring with credit risk-off.
Rate panic aggregate	$RatePanic_t = 0.5EZ(RR_t RL10_t VL_t) + 0.5EZ(RR_t CRO_t)$	The score's rate-panic component.
Credit panic	$CreditPanic_t = EZ(VS10_t \times CW_t)$	VIX spike with BAA–AAA spread widening.
Re-entry diagnostic	$ReentryOversold_t = EZ(DD_t \times VL_t)$	Oversold risky sleeve with high VIX.
Re-entry diagnostic	$EZ(VR5_t \times CA5_t)$	VIX relief with short-horizon credit appetite.
Re-entry diagnostic	$EZ(DD_t \times VR5_t)$	Drawdown depth followed by VIX relief.
Re-entry diagnostic	$EZ(VL_t \times VR5_t \times CA5_t)$	High VIX, VIX relief, and credit appetite together.
Re-entry diagnostic	$EZ(DD_t \times VR5_t \times CA5_t)$	Deep drawdown, VIX relief, and credit appetite together.
Re-entry aggregate	Average of the four relief products, then $ReentryScore_t = 0.8ReentryRelief_t + 0.2ReentryOversold_t$.	Used for diagnostics only; it does not create an additional trading threshold.

The primitive definitions are

$$RR_t = z(-\Delta TNX_{21,t}), \quad RL10_t = z(-RTrailing_{10,t}), \quad (30)$$

$$CRO_t = z(-HYGSHYRel_{21,t}), \quad CW_t = z(\Delta CS_{21,t}), \quad (31)$$

$$DD_t = z(-RDrawdown_t), \quad VR5_t = z(-\Delta VIX_{5,t}), \quad (32)$$

$$CA5_t = z(HYGTrailing_{5,t} - SHYTrailing_{5,t}). \quad (33)$$

Thus the V-shape score uses falling yields only when they coincide with equity loss, volatility stress, or credit stress; ordinary rate relief by itself is not sufficient to create a large cash allocation.

A.4 From Diagnostic Scores to Tradable Cash Weights

Both modules convert continuous scores into continuous raw cash weights, then apply the same economic materiality rule. For slow-tail, the current direct score combines cash-rising pressure, compressed-risk-premium states, rate headwind, and rate-path stress:

$$CashRaw_t^{slow} = MaxCash \cdot \sigma \left(\frac{SlowZ_t}{\tau} \right)^\gamma, \quad (34)$$

where $\sigma(\cdot)$ is the logistic function and γ is the cash convexity. For V-shape, the brake score is expanding-standardized and directly mapped into cash:

$$CashRaw_t^v = MaxCash \cdot \sigma \left(\frac{BrakeZ_t}{\tau} \right)^\gamma. \quad (35)$$

The shared rule is:

$$TargetCash_t = \begin{cases} CashRaw_t, & CashRaw_t \geq 0.30, \\ 0, & CashRaw_t < 0.30. \end{cases} \quad (36)$$

Entry to cash uses the target immediately; exit from cash uses the smoothing update

$$w_t^C = (1 - \eta_{exit})w_{t-1}^C + \eta_{exit}TargetCash_t, \quad \eta_{exit} = 0.25. \quad (37)$$

This is why the strategy can respond quickly during crash onset while avoiding excessive whipsaw when returning to the risky sleeve.

The first diagnostic layer is a screen, not a trading rule. The signs in Table 23 should not be read mechanically. Some stress variables have positive coefficients because high stress is often followed by recovery, while cash-rising and low-volatility variables point to weaker future $R - C$ compensation. This mixture is exactly why the paper separates slow-tail and V-shape modules instead of forcing all candidates into one monotone score.

Table 23: Continuous Main-Effect Screen for Future $R - C$

Variable	Role	63d Coef.	63d HAC t	Nonoverlap Coef.	Nonoverlap t	21d HAC t	126d HAC t
Cash yield high	cash compensation	-0.58	-0.42	0.47	0.20	-0.87	-0.10
Cash yield rising	cash compensation	-1.73	-2.43	1.29	0.47	-1.82	-1.20
Rate headwind	rate path	-1.61	-1.26	-5.79	-2.78	-0.28	-1.63
Curve inversion	rate path	0.32	0.31	1.76	0.94	-0.21	0.21
Risky drawdown	risk stress	2.92	2.30	3.52	2.27	2.62	1.82
High VIX	risk stress	4.79	2.39	7.80	2.42	1.82	1.75
VIX spike	risk stress	2.58	5.12	10.19	2.82	1.07	7.20
Credit risk-off	credit stress	2.80	6.05	4.78	2.40	2.29	4.46
Credit widening	credit stress	2.70	4.56	1.90	1.94	3.43	3.12
R strength	risk momentum	-1.91	-0.90	-1.58	-0.48	-0.93	0.28
R underwater	risk stress	3.55	1.89	3.25	1.60	2.31	1.12
Low volatility	calm regime	-8.98	-2.56	-11.60	-2.40	-2.59	-1.54

The main-effect screen leaves two broad families. The first is a low-compensation family: cash yield rising, low volatility, and risky-sleeve strength in calm markets tend to imply lower forward $R - C$. The second is a stress-reversal family: VIX and credit-stress variables often have positive forward coefficients because severe stress is followed by rebound. The slow-tail screen in Table 24 uses the first family, while the V-shape screen in Table 25 uses the second family as a crash-brake signal.

Table 24: Slow-Tail Main and Interaction Component Screen

Component	Role	63d Coef.	63d HAC t	Nonoverlap Coef.	Nonoverlap t	21d HAC t	126d HAC t
Cash rising	main effect	-0.25	-3.35	0.70	0.29	-2.74	-4.82
R strength x low vol	two-way	2.62	3.81	2.31	1.07	4.73	4.49
R strength x low vol x low VIX	three-way	-3.29	-7.06	-3.88	-1.77	-4.55	-7.40
R strength x low VIX	two-way	1.55	0.97	1.74	0.97	2.75	0.51
Cash high x R strength x low VIX	three-way	0.33	0.35	2.58	1.63	0.35	0.85
Cash high x R strength x low vol	three-way	0.05	0.07	4.67	1.96	0.23	0.22
Rate headwind	main effect	-0.66	-0.75	-3.68	-2.22	0.47	-1.22
Rate headwind x VIX spike	two-way	-1.04	-1.41	-0.87	-0.44	0.82	-1.22
Rate headwind x credit risk-off	two-way	-1.96	-2.75	-3.39	-1.27	-0.83	-2.61
Rate headwind x R underwater	two-way	-2.60	-2.09	-2.75	-1.52	-2.24	-2.84
Drawdown x rate headwind x credit risk-off	three-way	-2.52	-7.97	-6.76	-2.28	-1.59	-7.11
Slow direct score	aggregate score	-0.39	-3.50	-3.22	-1.40	-2.92	-3.30

The direct slow-tail score survives because the combined state is more informative than many isolated interactions. In Table 24, the aggregate score has a negative 63-day coefficient and a negative non-overlapping coefficient, meaning worse direct slow-tail states correspond to weaker future $R - C$ compensation. This aggregate evidence is more important than any single component because the policy trades on the combined compensation score, not on isolated interaction terms.

Table 25: V-Shape Brake and Re-Entry Component Screen

Component	Role	63d Coef.	63d HAC t	Nonoverlap Coef.	Nonoverlap t	21d HAC t
VIX level	brake main	1.37	1.73	4.89	2.68	1.49
VIX spike	brake main	0.50	1.41	-2.83	-1.70	-0.06
Rate panic	brake interaction	1.87	3.35	-5.57	-1.17	0.03
Credit panic	brake interaction	0.90	2.21	-2.95	-0.97	0.39
Reentry oversold	reentry diagnostic	1.92	1.07	8.73	2.69	1.66
Reentry relief	reentry diagnostic	2.66	2.93	10.80	2.17	0.41

The V-shape screen requires a different interpretation from the slow-tail screen. Table 25 is not a simple “high score means low future return” table. For crash brakes, high stress can precede positive forward $R - C$ because the crash has already occurred and recovery follows. The brake score is therefore evaluated by event loss reduction and drawdown control, not only by the sign of a 63-day predictive coefficient. The re-entry rows are kept as diagnostics because they describe rebound conditions but are not used as an additional hard trading trigger in the final material policy.

Table 26: Slow-Tail Entry and Active-State Diagnostics

Bucket	Days	Future 21d $R - C$	Future 63d $R - C$	Future 126d $R - C$	63d Win Rate	Avg. Raw Cash	Avg. Cash
entry days	17	-0.94%	-1.45%	-1.39%	41.18%	31.79%	8.29%
material active days	93	-2.33%	-2.09%	-0.15%	38.71%	34.26%	30.15%
cash raw above 30pct	92	-2.25%	-2.11%	-0.17%	38.04%	34.31%	30.00%
actual cash weight days	202	-0.27%	0.00%	2.35%	49.50%	30.09%	21.39%

Slow-tail activation is economically sparse. Table 26 reports 17 entry days and 93 material active days in the main window. The future 63-day $R - C$ mean is negative for the material active state, which confirms that the cash allocation is concentrated in the intended weak-compensation regime. The broader actual-cash-weight row includes smoothed exits, so its future $R - C$ average is less negative; this is expected because recovery days are included during the exit path.

Table 27: Slow-Tail Cost Sensitivity

Cost bp	CAGR	Sharpe	Max DD	Ann. Turnover	Avg. Cash
0	17.96%	0.94	-33.59%	108.36%	2.19%
5	17.89%	0.94	-33.59%	108.36%	2.19%
10	17.83%	0.94	-33.59%	108.36%	2.19%
20	17.70%	0.93	-33.59%	108.36%	2.19%

Transaction costs do not materially alter the slow-tail conclusion over the tested range. As reported in Table 27, annual turnover is modest compared with the V-shape module, so moving from 0bp to 20bp reduces CAGR only gradually. This matters because the slow-tail module is meant to handle multi-week or multi-month deterioration, not high-frequency switching.

Table 28: Slow-Tail Fixed Holdout Check

Period	Role	CAGR	Max DD	Ann. Turnover	Avg. Cash	100% R CAGR	100% R Max DD
2018–2021 train	fixed direct_slow_157	21.98%	-33.59%	17.12%	0.13%	22.08%	-33.59%
2022–2026 holdout	fixed direct_slow_157	13.72%	-19.66%	195.54%	4.51%	12.47%	-23.78%

The fixed train/holdout check is deliberately simple. In Table 28, a rule selected on the 2018–2021 training period is almost inactive during that training period, but it improves both CAGR and drawdown in the 2022–2026 holdout window. This supports the interpretation that the slow-tail state becomes valuable only once cash yield and risky-sleeve compensation conditions change after 2021.

Table 29: V-Shape Score Sorting Diagnostics

Score	Bucket	Days	Future 63d $R - C$ Mean	63d Win Rate
brake z	Q1	430	2.31%	68.37%
brake z	Q2	429	2.69%	75.06%
brake z	Q3	429	2.80%	71.33%
brake z	Q4	429	5.04%	80.89%
brake z	Q5	429	4.19%	71.33%
reentry z	Q1	431	4.46%	81.67%
reentry z	Q2	430	2.79%	72.09%
reentry z	Q3	430	2.25%	68.37%
reentry z	Q4	430	1.96%	64.42%
reentry z	Q5	430	5.58%	80.70%

The score-sorting exercise is diagnostic rather than a portfolio rule. The brake score does not need monotone Q1–Q5 return sorting because it is not intended to forecast ordinary returns across all states (Table 29). Its purpose is to identify high-stress tails where temporary cash has drawdown value. The re-entry score has a high Q5 future mean, which is consistent with rebound conditions, but it remains diagnostic to avoid adding a second discretionary threshold.

Table 30: V-Shape Cost Sensitivity

Cost bp	CAGR	Sharpe	Max DD	Ann. Turnover	Avg. Cash
0	17.94%	1.10	-23.77%	532.41%	3.65%
5	17.63%	1.08	-23.77%	532.41%	3.65%
10	17.32%	1.07	-23.77%	532.41%	3.65%
20	16.69%	1.03	-23.78%	532.41%	3.65%

The V-shape module is more cost-sensitive because it trades more. Table 30 shows a stronger cost effect than Table 27. Even at 20bp, however, maximum drawdown remains materially below 100% R . This reinforces the interpretation that V-shape is a risk-control component whose economic case comes primarily from drawdown reduction.

Table 31: V-Shape Yearly Breakdown Versus 100% R

Year	V-shape CAGR	V-shape Max DD	Avg. Cash	100% R CAGR	100% R Max DD
2017	23.08%	-1.88%	5.59%	27.03%	-2.08%
2018	-2.96%	-17.00%	7.68%	-3.03%	-19.71%
2019	32.32%	-8.03%	0.57%	32.90%	-8.03%
2020	43.96%	-15.19%	7.64%	23.71%	-33.59%
2021	29.24%	-4.92%	0.84%	31.04%	-4.92%
2022	-17.12%	-23.77%	0.00%	-17.12%	-23.78%
2023	28.40%	-9.73%	0.00%	28.40%	-9.73%
2024	18.21%	-8.00%	3.28%	22.51%	-8.88%
2025	15.37%	-14.82%	7.02%	17.46%	-19.66%
2026	20.94%	-8.02%	7.61%	28.88%	-7.95%

The yearly decomposition clarifies where the V-shape filter earns its keep. In Table 31, the module’s major positive contribution is 2020, where it reduces the crash drawdown

and then still participates enough in the rebound to outperform the risky sleeve for the year. It does not help in 2022 because the 2022 regime is not a fast V-shaped crash. In 2024–2026, the module gives up some return in exchange for lower drawdowns during stress episodes.

Table 32: Complete OOS Return Matrix from Sub-Strategy OOS Weights

OOS Window	Mode	Method	CAGR	Max DD	Ann. Turnover	Avg. Cash	CAGR Diff	MDD Improvement
main-oos	Expanding	Max-cash	19.35%	-22.05%	305.09%	5.22%	1.76%	11.53%
main-oos	Expanding	Slow-tail	18.79%	-33.59%	114.57%	2.44%	1.20%	0.00%
main-oos	Expanding	V-shape	17.98%	-25.82%	202.58%	2.92%	0.39%	7.77%
main-oos	Expanding	100% R	17.59%	-33.59%	0.00%	0.00%	0.00%	0.00%
main-oos	Rolling	Max-cash	18.50%	-22.05%	351.81%	6.15%	0.90%	11.53%
main-oos	Rolling	Slow-tail	18.12%	-33.59%	138.31%	2.77%	0.53%	0.00%
main-oos	Rolling	V-shape	17.22%	-26.18%	246.39%	3.73%	-0.37%	7.40%
main-oos	Rolling	100% R	17.59%	-33.59%	0.00%	0.00%	0.00%	0.00%
post2022-oos	Expanding	Max-cash	20.76%	-15.53%	397.21%	4.56%	-2.82%	4.13%
post2022-oos	Expanding	Slow-tail	21.19%	-19.74%	169.65%	1.62%	-2.39%	-0.08%
post2022-oos	Expanding	V-shape	21.90%	-14.75%	274.19%	3.38%	-1.67%	4.92%
post2022-oos	Expanding	100% R	23.58%	-19.66%	0.00%	0.00%	0.00%	0.00%
post2022-oos	Rolling	Max-cash	18.91%	-17.26%	470.06%	6.24%	-4.67%	2.41%
post2022-oos	Rolling	Slow-tail	21.19%	-19.74%	169.65%	1.62%	-2.39%	-0.08%
post2022-oos	Rolling	V-shape	19.35%	-16.88%	370.97%	5.24%	-4.23%	2.78%
post2022-oos	Rolling	100% R	23.58%	-19.66%	0.00%	0.00%	0.00%	0.00%

The most compact audit of the combined walk-forward experiment is the full OOS return matrix. Table 32 reports component returns and combined returns on the same OOS windows. The main OOS window supports the combination: max-cash improves both CAGR and drawdown relative to 100% R . The post-2022 rows show the limitation of a simple upper-envelope rule: it reduces drawdown relative to 100% R , but it can underperform the best standalone component and the risky sleeve on CAGR when the rebound is strong. The comparison clarifies the combination objective: improve defensive balance, not maximize standalone component CAGR in every subperiod.

Table 33: Walk-Forward Parameter Selection Diagnostics

Module	OOS Window	Mode	Blocks	Unique Configs	Test Window	Top Config Blocks	Top Share	Median Train CAGR
Slow-tail	main-oos	Expanding	28	4	2019-07-01–2026-04-30	16	57.14%	–
Slow-tail	main-oos	Rolling	28	10	2019-07-01–2026-04-30	7	25.00%	–
Slow-tail	post2022-oos	Expanding	14	6	2023-01-04–2026-04-30	5	35.71%	–
Slow-tail	post2022-oos	Rolling	14	6	2023-01-04–2026-04-30	5	35.71%	–
V-shape	main-oos	Expanding	31	8	2018-09-27–2026-04-30	14	45.16%	17.22%
V-shape	main-oos	Rolling	31	20	2018-09-27–2026-04-30	4	12.90%	16.89%
V-shape	post2022-oos	Expanding	18	4	2022-01-03–2026-04-30	14	77.78%	17.14%
V-shape	post2022-oos	Rolling	18	13	2022-01-03–2026-04-30	5	27.78%	17.52%

Parameter re-selection is substantial but structured. Table 33 shows that expanding selection concentrates more heavily in a few top configurations, while rolling selection uses more unique configurations, especially for V-shape. The direct slow-tail selection files record the chosen configuration by block but do not store the training-period performance metrics, so the median training CAGR cells are left blank for that module. The configuration-count evidence is still useful: slow-tail is more concentrated under expanding selection, whereas V-shape remains deliberately more adaptive.

B Start-Date Sensitivity Selection Stability

Portfolio performance and parameter stability tell different parts of the validation story. The performance tables above summarize realized results across requested starts, while

Table 34 reports the parameter-selection layer behind those results. The slow-tail module is more stable under expanding selection, while the V-shape module is deliberately more adaptive under rolling selection. This is expected: the V-shape filter is a crash brake whose preferred VIX, rate, and credit-panic weights can shift after recent fast-crash episodes. As in Table 33, the direct slow-tail selection artifacts record block-level configuration choices but not training-period performance columns, so those training-metric cells are intentionally left blank.

Table 34: Start-Date Sensitivity: Component Selection Stability Across Requested Starts

Component	Start	Mode	Blocks	Unique	Top Share	Med. Train CAGR	Med. Train Sharpe
Slow-tail	2018-06-28	Expanding	28	4	57.14%	—	—
V-shape	2018-06-28	Expanding	31	8	45.16%	17.22%	1.05
Slow-tail	2018-06-28	Rolling	28	8	25.00%	—	—
V-shape	2018-06-28	Rolling	31	20	12.90%	16.89%	1.02
Slow-tail	2018-07-30	Expanding	27	5	59.26%	—	—
V-shape	2018-07-30	Expanding	31	9	45.16%	17.38%	1.06
Slow-tail	2018-07-30	Rolling	27	11	37.04%	—	—
V-shape	2018-07-30	Rolling	31	21	12.90%	16.67%	1.09
Slow-tail	2018-08-29	Expanding	27	4	62.96%	—	—
V-shape	2018-08-29	Expanding	31	9	41.94%	17.21%	1.04
Slow-tail	2018-08-29	Rolling	27	9	37.04%	—	—
V-shape	2018-08-29	Rolling	31	18	12.90%	16.11%	1.04
Slow-tail	2018-10-29	Expanding	26	4	65.38%	—	—
V-shape	2018-10-29	Expanding	30	7	50.00%	17.24%	1.05
Slow-tail	2018-10-29	Rolling	26	9	42.31%	—	—
V-shape	2018-10-29	Rolling	30	19	16.67%	16.56%	1.04
Slow-tail	2019-01-02	Expanding	26	4	65.38%	—	—
V-shape	2019-01-02	Expanding	30	6	46.67%	17.06%	1.05
Slow-tail	2019-01-02	Rolling	26	8	34.62%	—	—
V-shape	2019-01-02	Rolling	30	18	13.33%	16.85%	1.09
Slow-tail	2020-01-02	Expanding	22	3	77.27%	—	—
V-shape	2020-01-02	Expanding	26	5	53.85%	17.36%	1.05
Slow-tail	2020-01-02	Rolling	22	8	40.91%	—	—
V-shape	2020-01-02	Rolling	26	17	15.38%	18.24%	1.15
Slow-tail	2022-01-03	Expanding	14	6	35.71%	—	—
V-shape	2022-01-03	Expanding	18	4	77.78%	17.14%	1.01
Slow-tail	2022-01-03	Rolling	14	7	35.71%	—	—
V-shape	2022-01-03	Rolling	18	13	27.78%	17.52%	0.94

The seven requested starts tell the same selection-stability story. In Table 34, the slow-tail expanding selector is relatively concentrated, especially for pre-2022 requested starts, while V-shape rolling selection remains dispersed across many configurations. This supports the paper’s interpretation of the two modules: slow-tail is a persistent-regime filter, whereas V-shape is a more event-sensitive brake.

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